UI BUILDER

AUTOMATIC CODE GENERATION TOOL TO DESIGN AND DEVELOP UI

Project ID: 2022-070



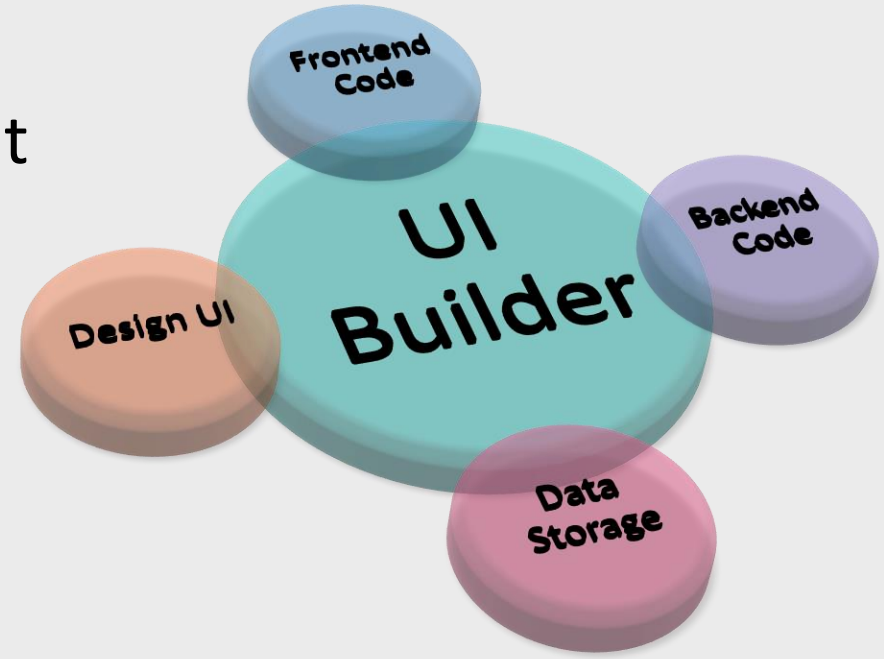
SLIIT
FACULTY OF COMPUTING

TEAM MEMBERS

Group ID: 2022-070	
Supervisor: Dr. Nuwan Kodagoda	
Co Supervisor: Ms. Kushnara Suriyawansa	
Student Name	**Student ID**
Ekanayake K.L.W (Group Leader)	IT19150758
Sakthivel.S	IT19158228
H.M. Kasuni Navodya	IT19144986
D.P Kavindi Gimshani	IT19150826

INTRODUCTION

- What is Automatic Code generation (ACG)?
- What is the impact on software development process with ACG?
- Why do we need a new tool for ACG?



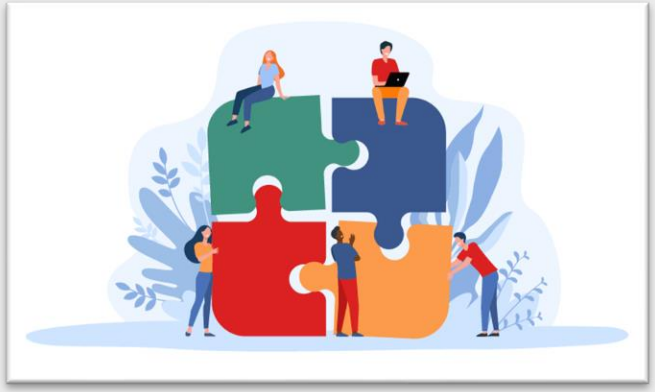
RESEARCH PROBLEM

- The main problem in the research is the UI designers, software engineers are unable to design the UI templates according to their preference.
- The users are unable to preview a live display of the UI templates.
- Designers and software engineers need to spend more effort & time to finalize UI preferred by end user.
- Non-technical people cannot use the UI designing tools currently in the market.
- Unable to develop front end code for the both mobile and web UI templates.
- Unable to develop the backend code along with the database from current tools in the market.

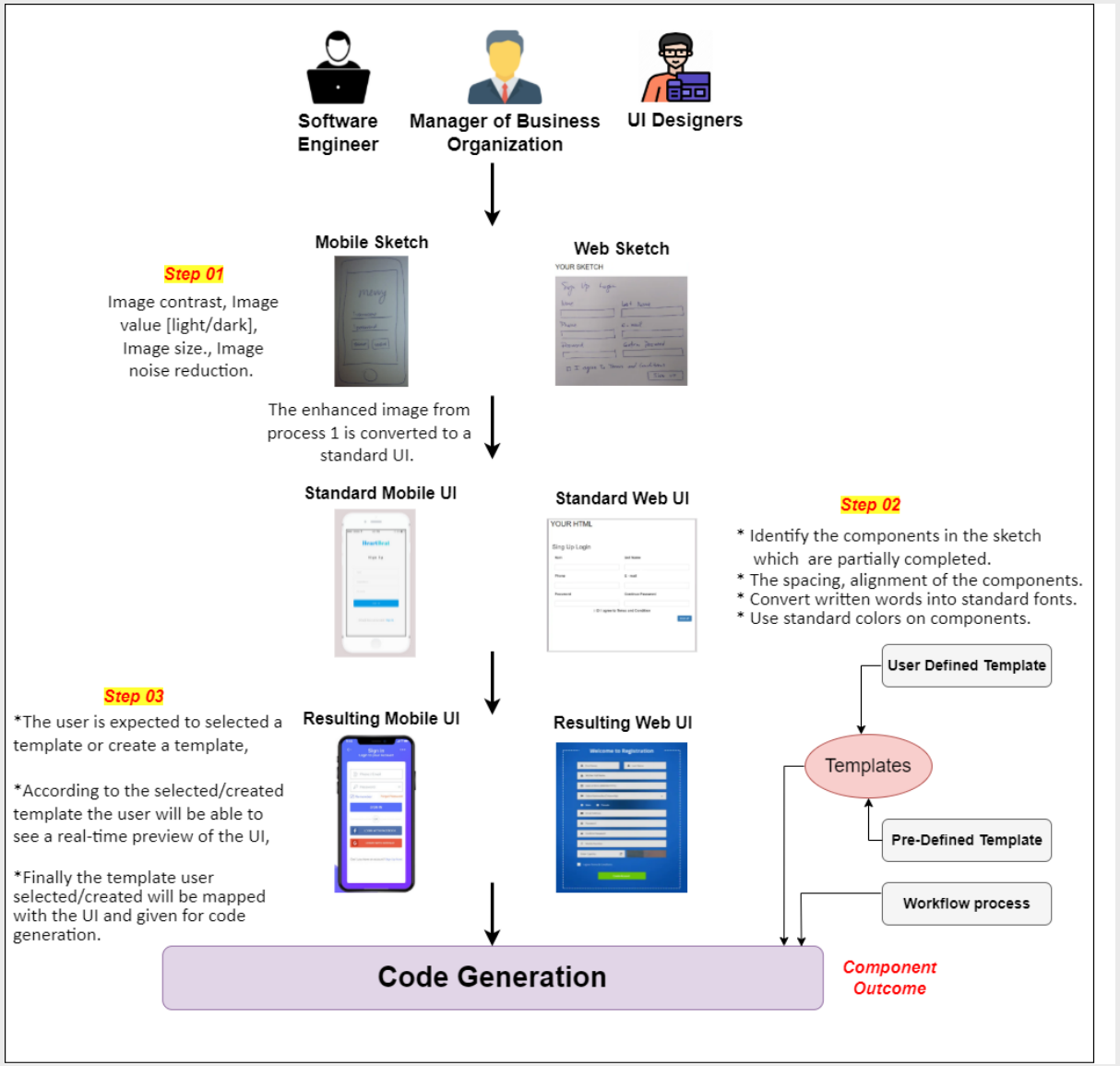


RESEARCH OBJECTIVES

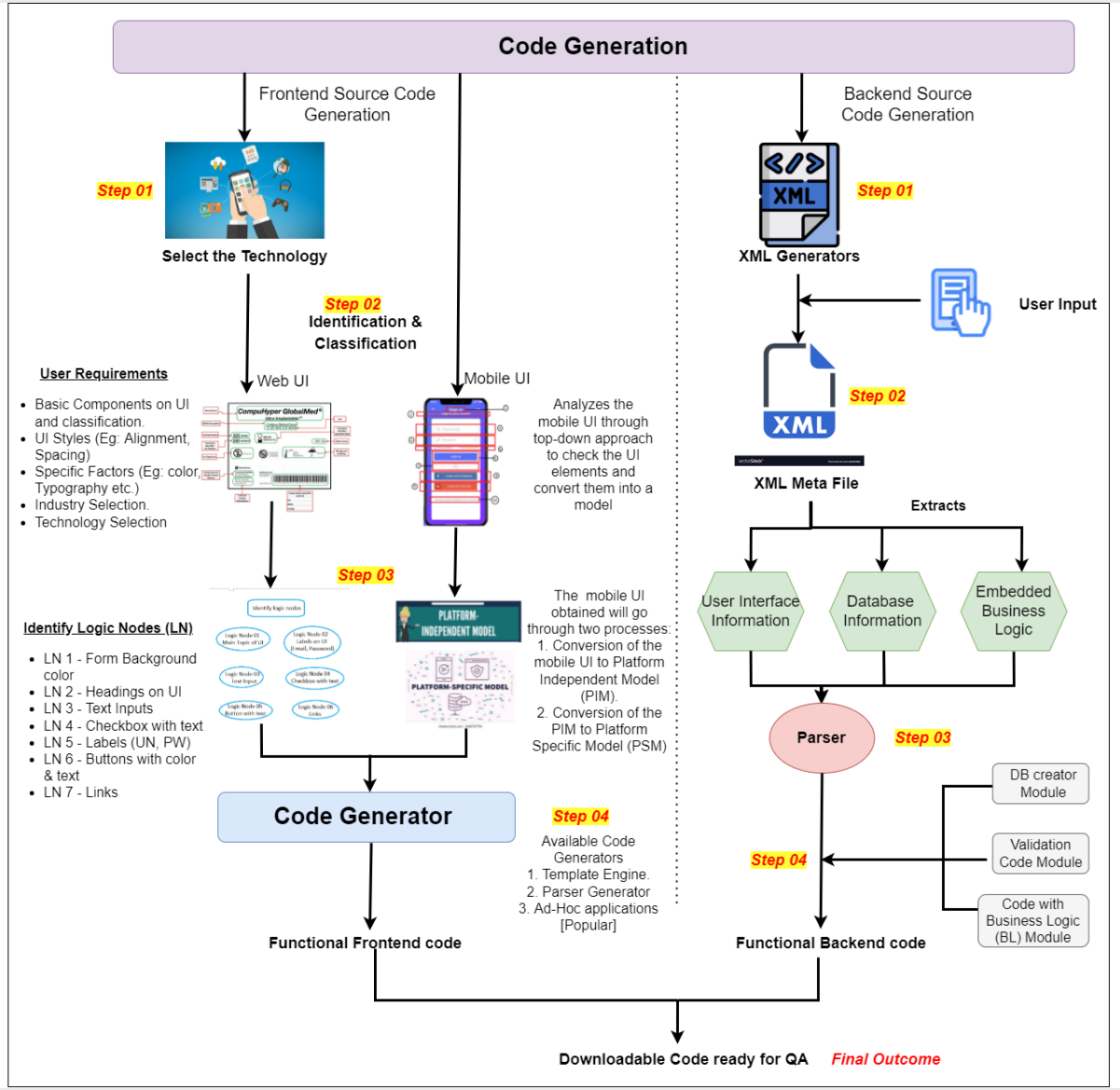
- Generate UI templates according to UI rough sketches given by users.
- Generate web frontend code from UI template using Logic Node concept.
- Generate mobile frontend code from UI template using Model Driven Architecture.
- Develop backend code and DB generator using forward engineering process.



SYSTEM DIAGRAM (PART 01)



SYSTEM DIAGRAM (PART 02)



OUR PLAN TO PROMOTE THE TOOL



- Conducting a live demonstration of code generation process in developer environment.
- Understanding the need and expectations of developers through automated feedback services integrated with the tool.
- Select startup companies and describe them the capabilities and advantages of using UI builder tool for their software development process.
- Use social media and publish an official website for the tool to make it reach out for wide range of people.
- Conduct a workshop on the quick setup of the tool in their PC for 1st and 2nd year IT students.
- Speak around ‘Anyone Can Code’ concept to make It stand unique among the other code generation tools.
- Use quick startup guides and YouTube tutorials for better usability and user-friendliness.



IT19150758|EKANAYAKE K.L.W

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY

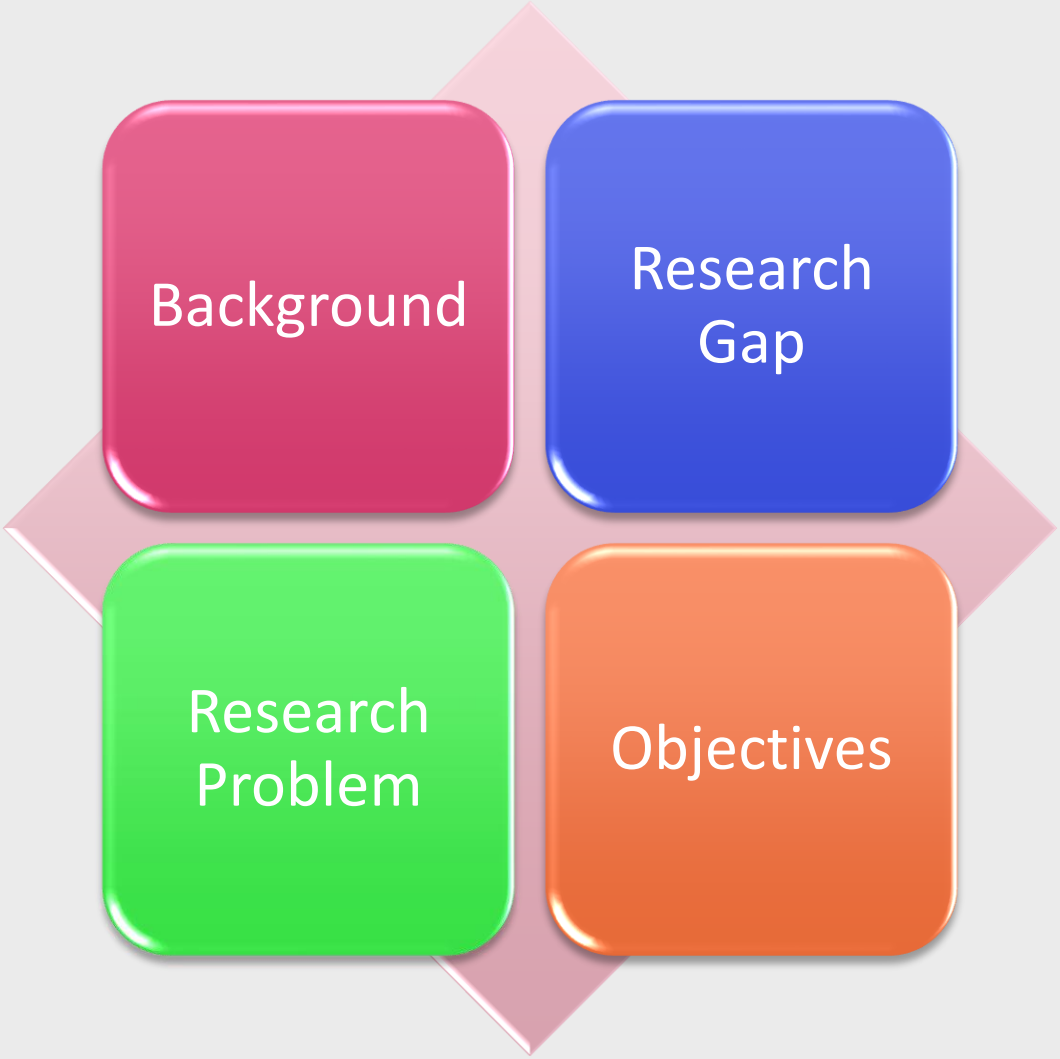
(SPECIALIZATION IN SOFTWARE ENGINEERING)

RESEARCH COMPONENT

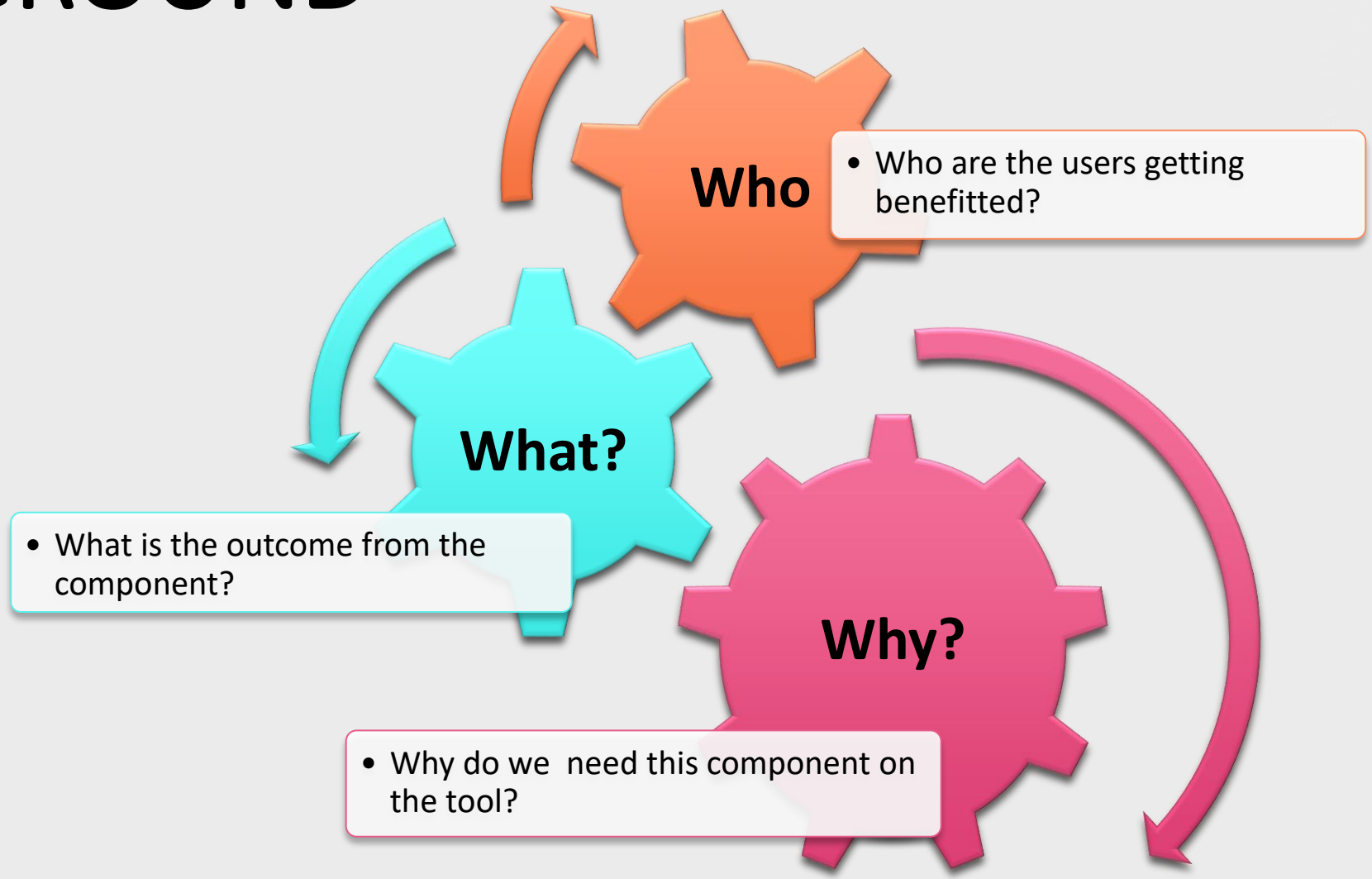
GENERATING UI TEMPLATES ACCORDING TO ROUGH SKETCHES.



INTRODUCTION



BACKGROUND


Who
• Who are the users getting benefitted?
What?
• What is the outcome from the component?
Why?
• Why do we need this component on the tool?

RESEARCH GAP

Features	Tools				
	Pix2Code	Sketch2Code	Adobe Comp	Sketch	'UI Builder'
Quick and easy	✓	✓	✗	✗	✓
Can create UI templates according to the rough sketches.	✗	✗	✗	✗	✓
User-friendliness	✓	✓	✗	✗	✓
Accuracy	✓	✓	✗	✗	✓
Can get user preferred UI templates	✗	✗	✗	✗	✓

RESEARCH PROBLEM

- How to generate UI template according to the rough sketch ?
- How identify the UI elements in the rough sketch ?
- How to align and design the UI template ?
- How to provide a better UI for the users.
- How is the image processing part working ?

OBJECTIVES

Main Objective

* The main objective in this research partition to generate a UI template according to a rough sketch.

Sub Objective 1

- Identifying clearly the drawings in the rough sheet.

Sub Objective 2

- Generate the UI template according to the rough sheet sketch.

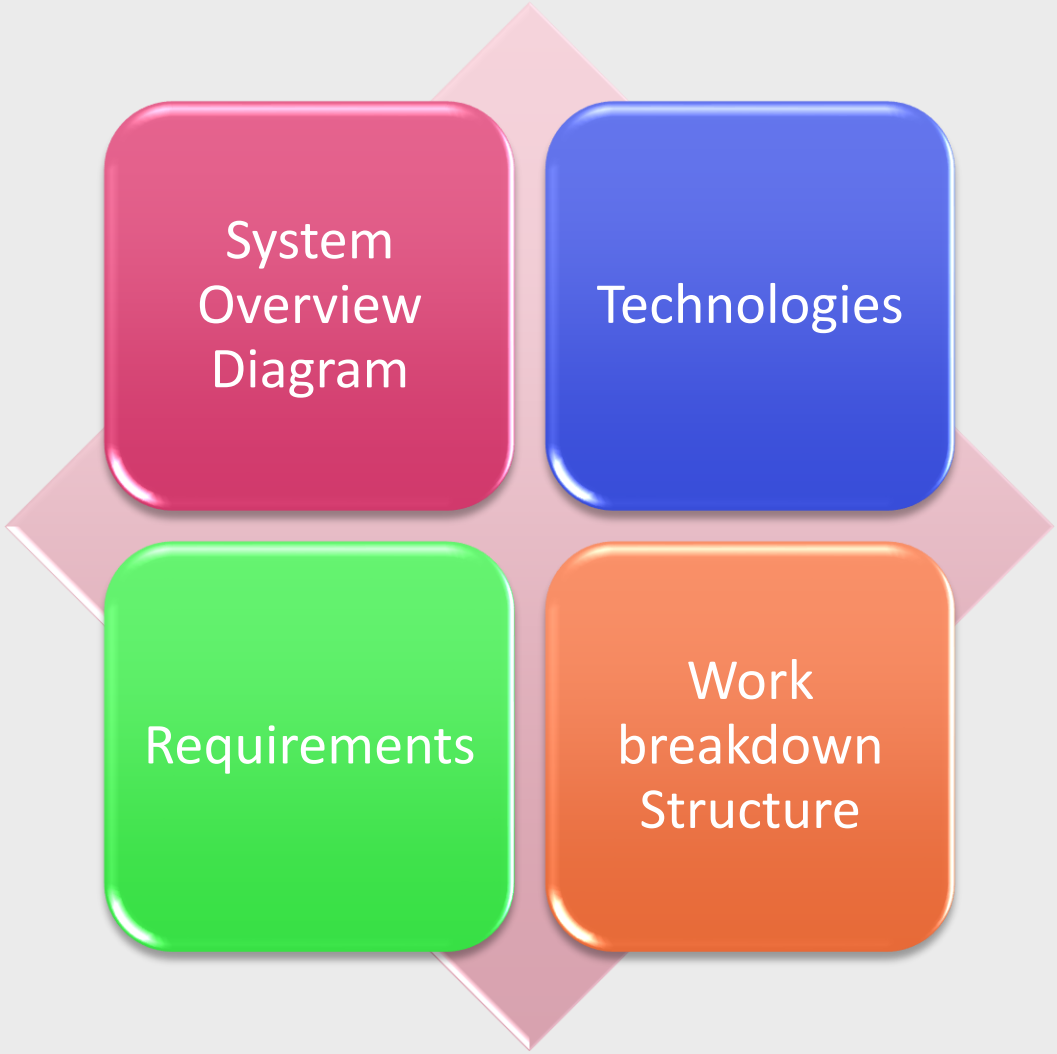
Sub Objective 3

- Adding colors, alignment and other designs.

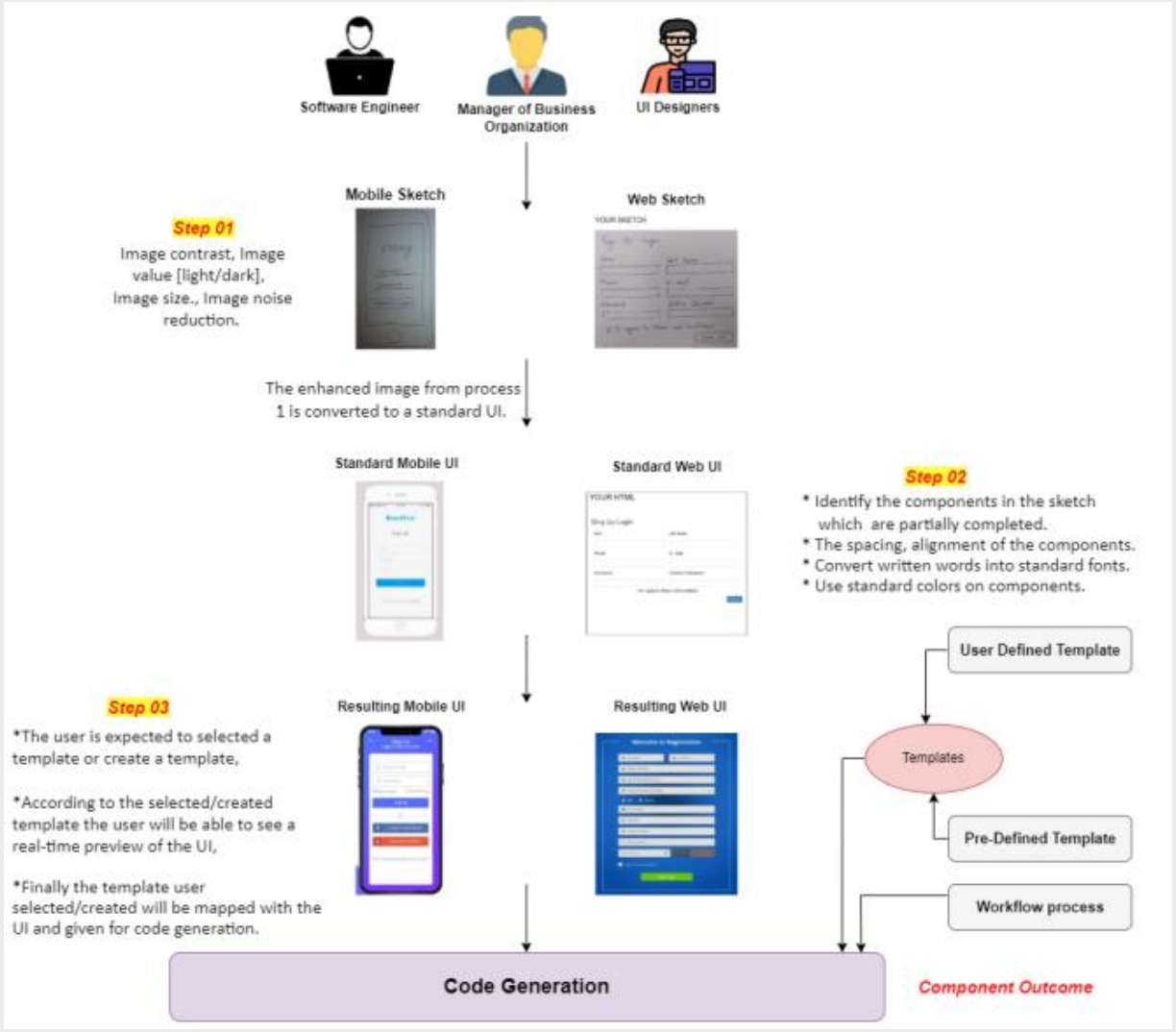
Sub Objective 4

- Design user preferred templates.

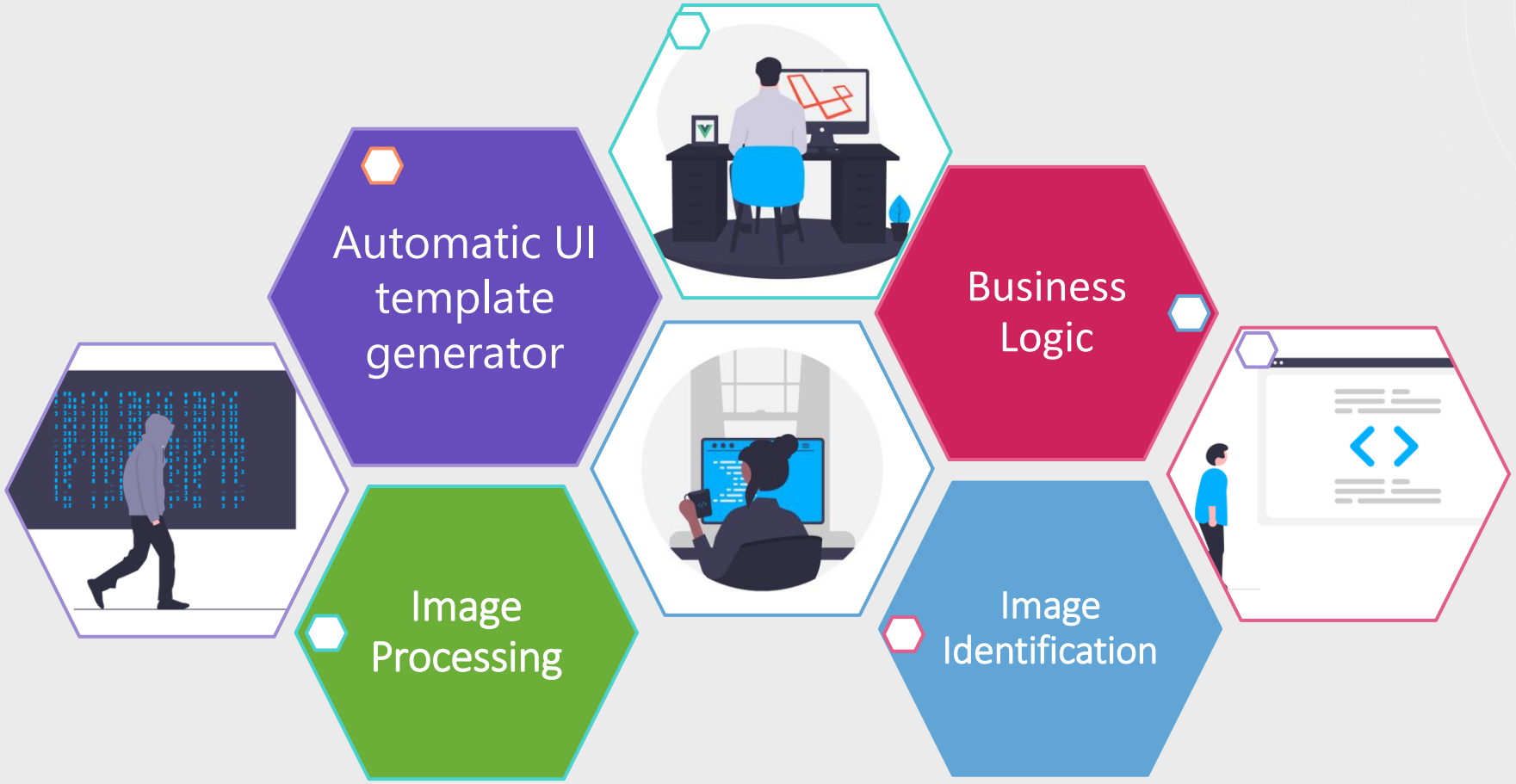
METHODOLOGY



SYSTEM OVERVIEW DIAGRAM



KEY PILLARS


Automatic UI template generator
Business Logic
Image Processing
Image Identification

TOOLS & TECHNOLOGIES

Technologies

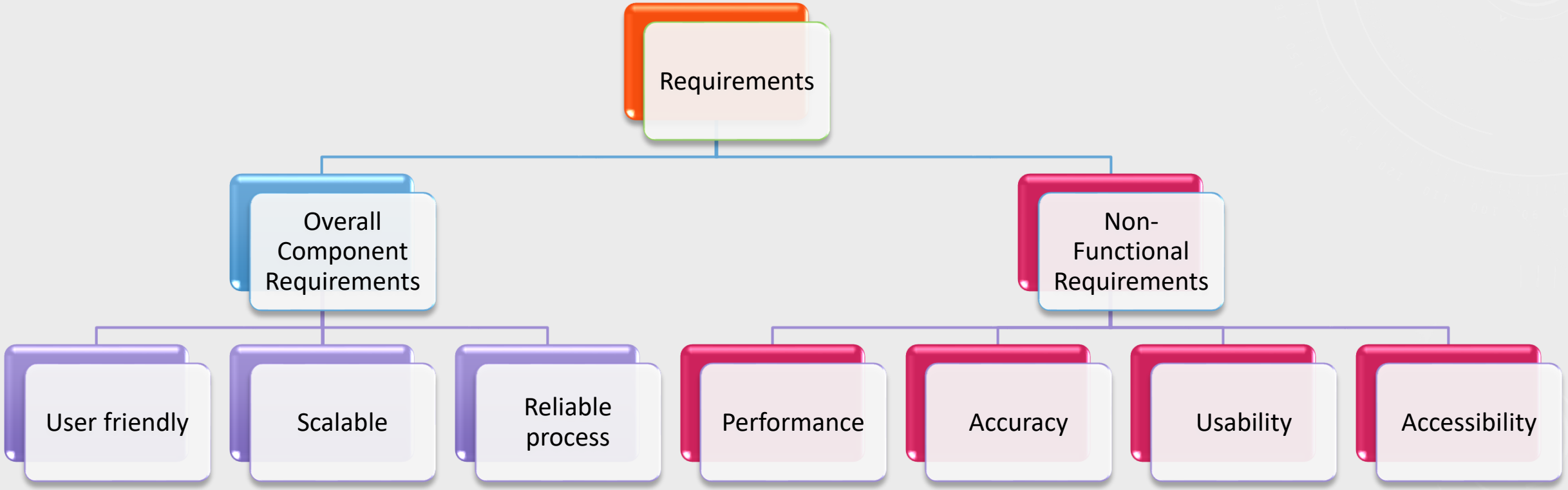
- Image Understanding and Processing(IUP)

Tools

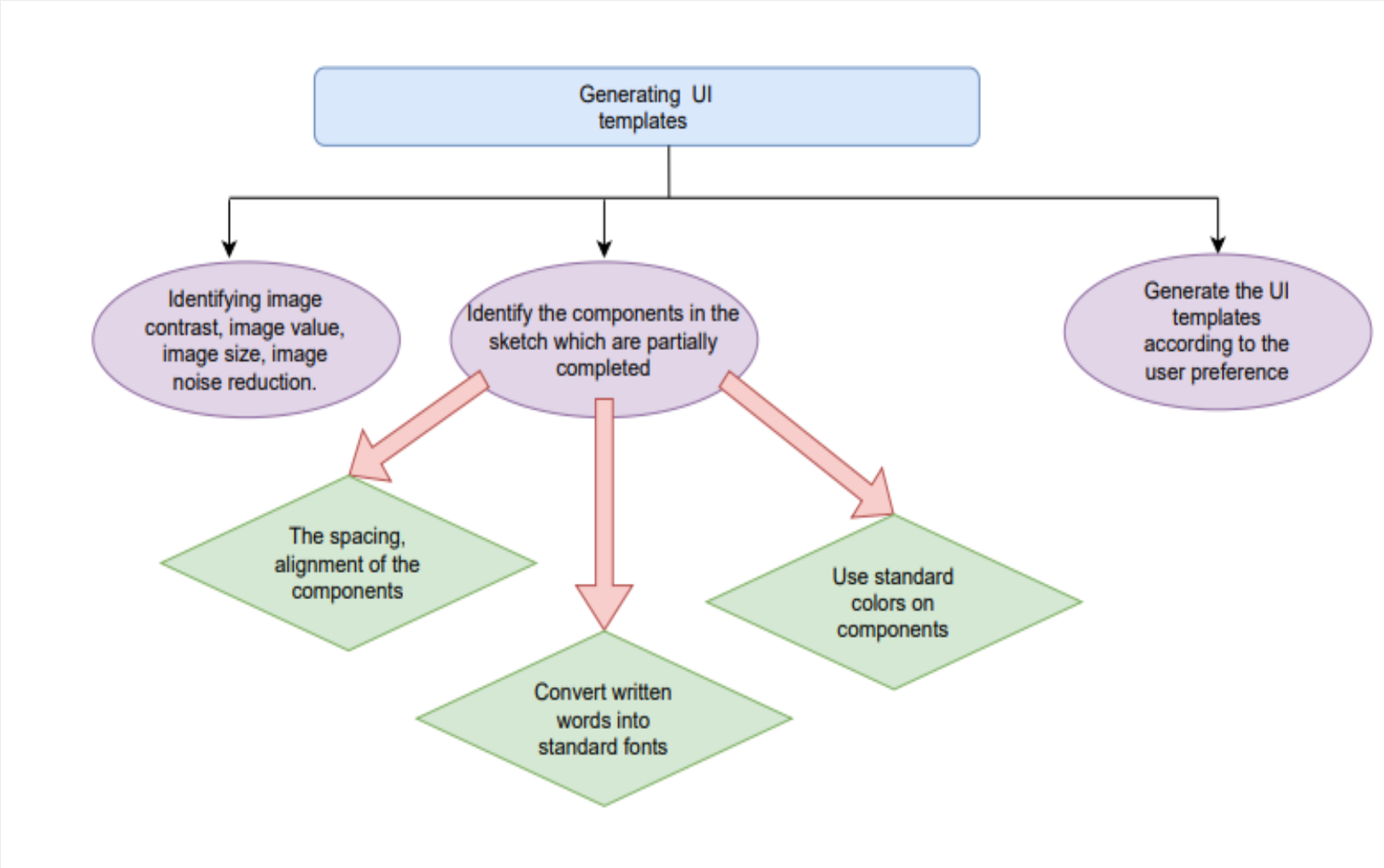
- Version Management – GitLab
- Project Management – Trello, OneNote



REQUIREMENTS



WORK BREAKDOWN STRUCTURE



GANTT CHART

Task Name	Timeline												
Description	Priority	January	February	March	April	May	June	July	August	September	October	November	December
Project Initiation													
Topic Evaluation	HIGH												
Project Charter	HIGH												
Project Cover sheet	MEDIUM												
Project Proposal (Draft)	MEDIUM												
Conduct a online survey - PHASE 01	HIGH												
Project proposal presentation	HIGH												
Project Proposal - Final report	HIGH												
Organizing and Requirement Anal													
System planning	HIGH												
Collecting required data - PHASE 02	MEDIUM												
Technology & algorithm analysis and selection	HIGH												
Writing Research paper	HIGH												
Implementation													
Identifying image specifications.	HIGH												
Isolate partially completed UI components	HIGH												
Generate UI Templates according to user preferences.	HIGH												
Experimental Analysis.	HIGH												
Testing													
Developer Testing	HIGH												
Final Report	HIGH												
Final Evaluation	HIGH												



IT19158228|SAKTHIVEL.S

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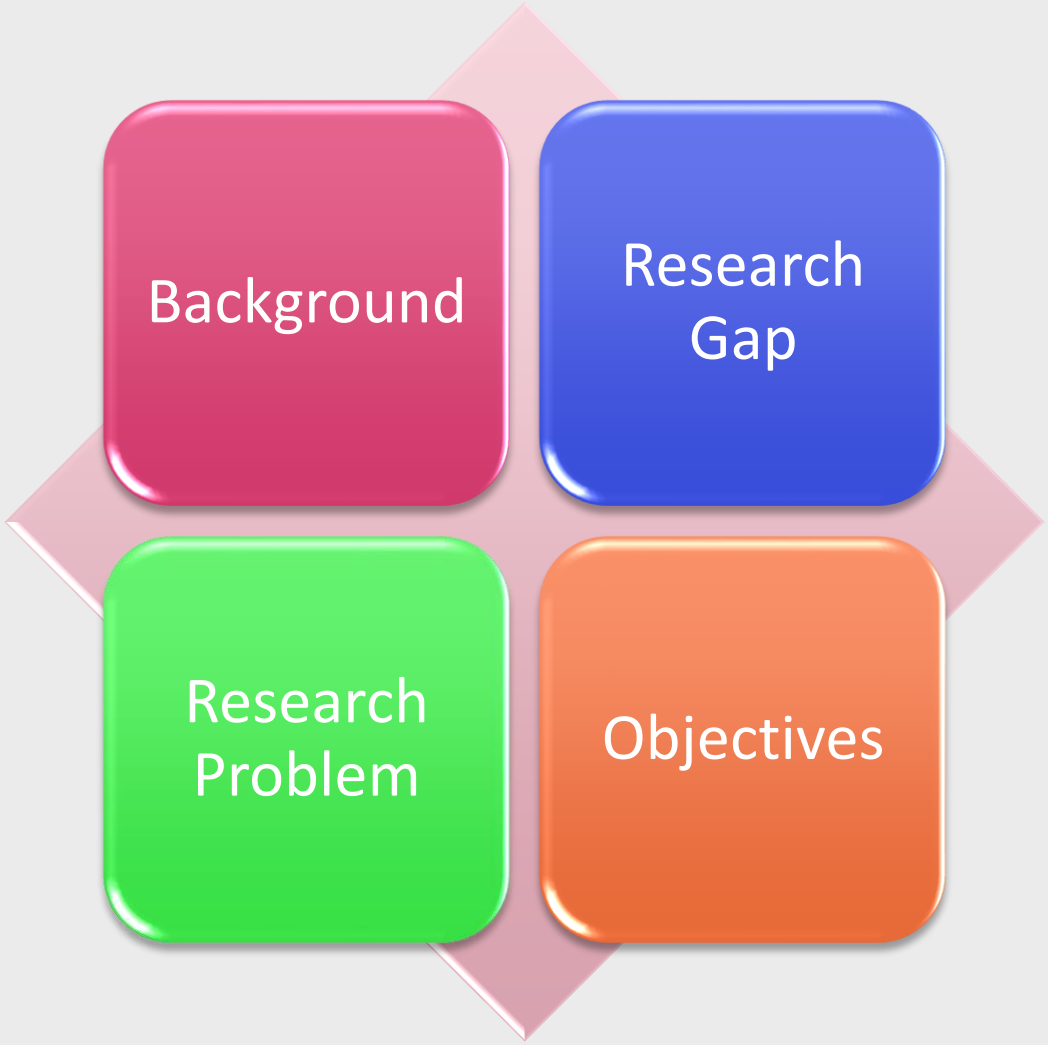
(SPECIALIZATION IN SOFTWARE ENGINEERING)

RESEARCH COMPONENT

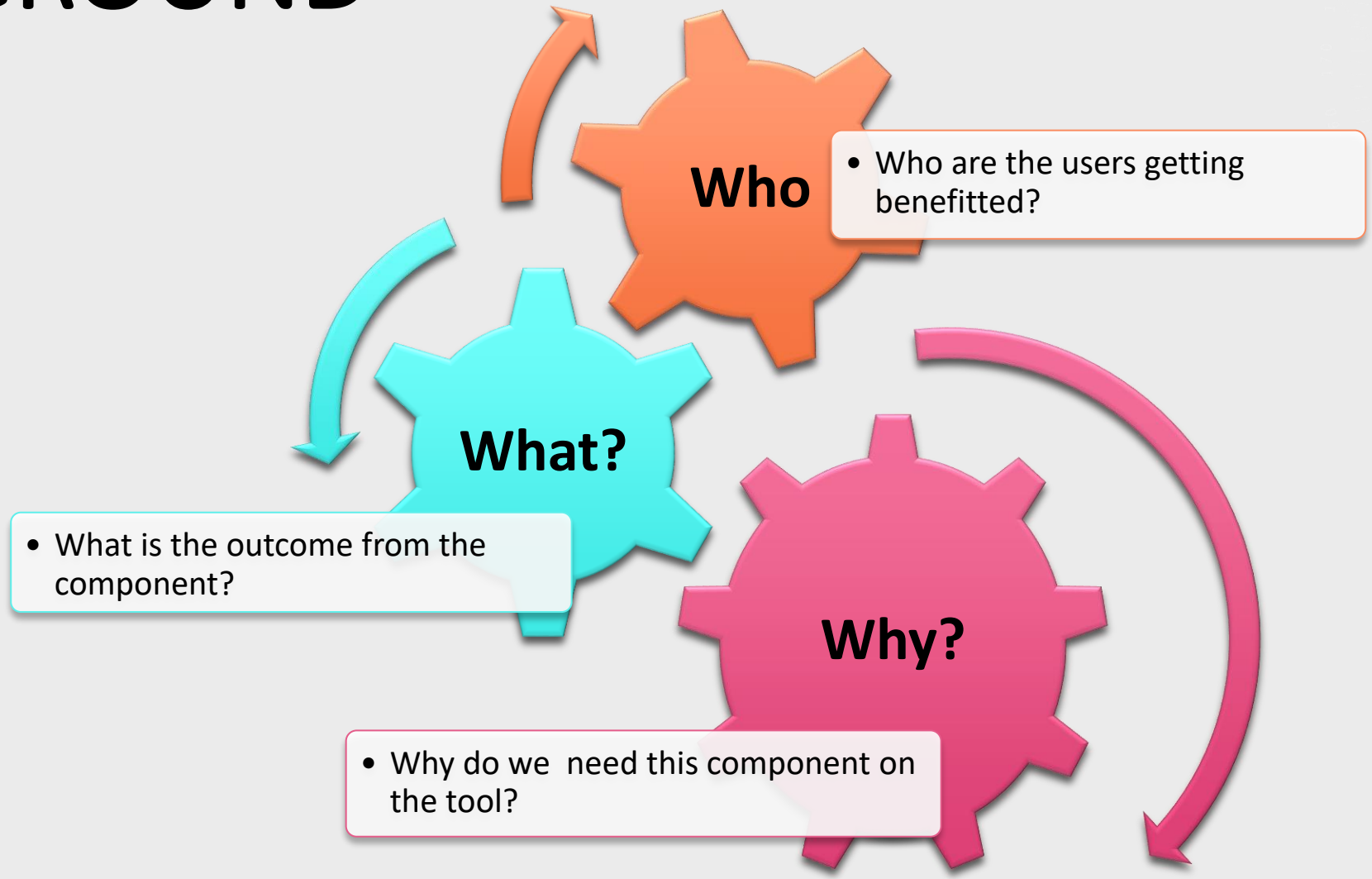
BACKEND CODE AND DATABASE GENERATOR



INTRODUCTION



BACKGROUND


Who
• Who are the users getting benefitted?
What?
• What is the outcome from the component?
Why?
• Why do we need this component on the tool?

RESEARCH GAP

Features & Aspects	Tools				
	Pix2Code	Sketch2Code	Rapid Project Builder	App Sheet	'UI Builder'
Easily understandable code generation process.	✓	✓	✗	✗	✓
BL Embedded backend code.	✗	✗	✓	✗	✓
Manageable database with validations.	✗	✗	✗	✗	✓
Coding best practices and code standards.	✓	✗	✗	✗	✓
Usability & applicability.	✗	✗	✗	✗	✓
Accuracy .	✓	✓	✗	✓	✓

RESEARCH PROBLEM



How to auto-generate business logic embedded backend code with no-developer involvement?



How to provide an easily manageable database with proper validations to preserve data integrity?



How to provide texture of real handwritten code on the auto-generated backend code?

OBJECTIVES

Main Objective

❖ Generate a fully functional Business Logic embedded backend code with database connectivity.

Sub Objective 1

• Understandable code generation process and continuous integration pipeline

Sub Objective 2

• Meet high code quality and standards with the generated code

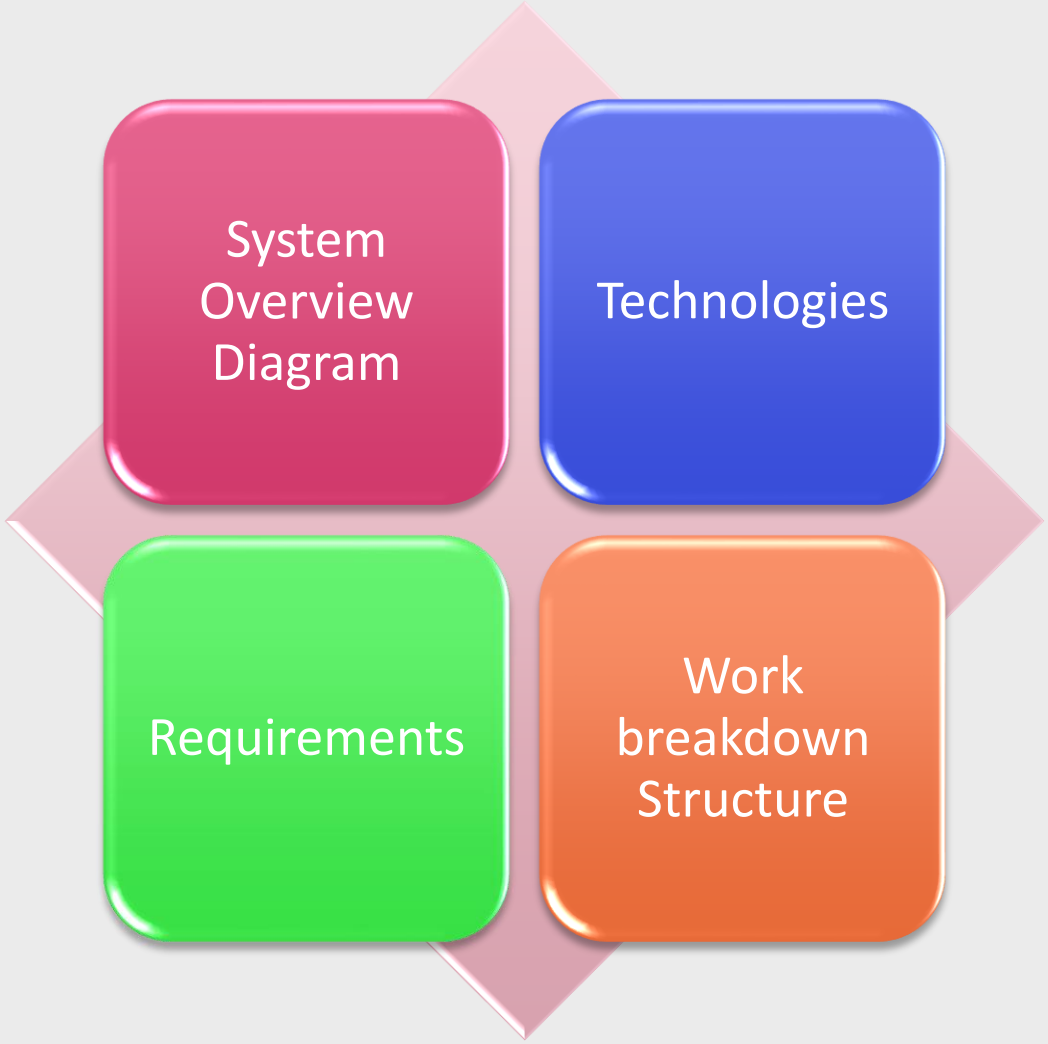
Sub Objective 3

• Encourage developers to use on their real production environment.

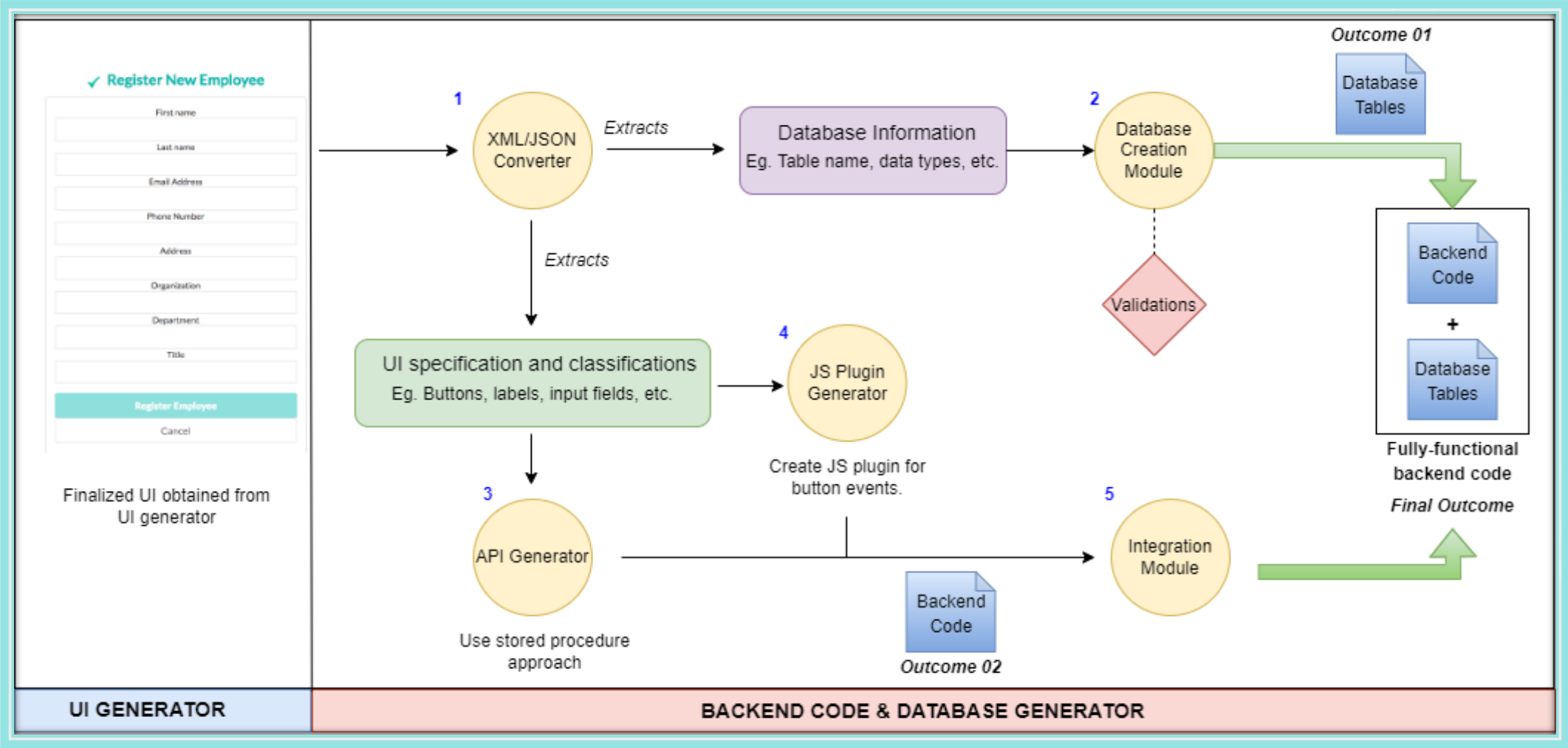
Sub Objective 4

• Code reusability and code refactoring.

METHODOLOGY



SYSTEM OVERVIEW DIAGRAM



KEY PILLARS

Combine all lower modules to complete the process.

Integration Module.

Create JS File for UI components with user actions.

JS Plugin Builder.

Create API for identified business operations

API Generator.

Create Database Tables with validations.

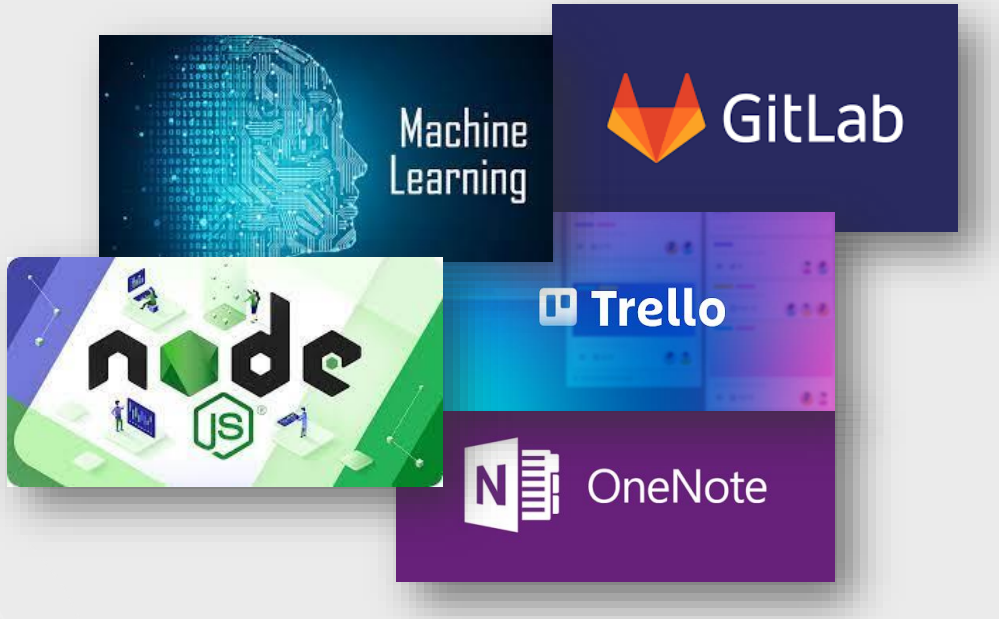
Automatic Database Creator.

Convert UI to XML Meta-File

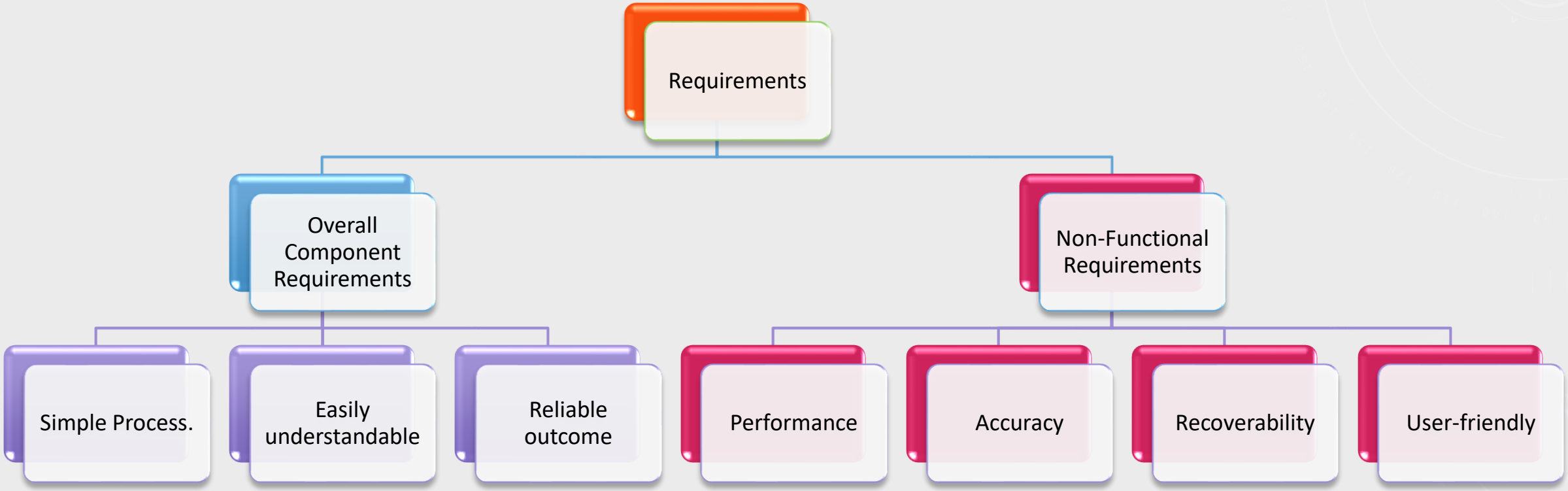
UI to XML Conversion Module.

TOOLS & TECHNOLOGIES

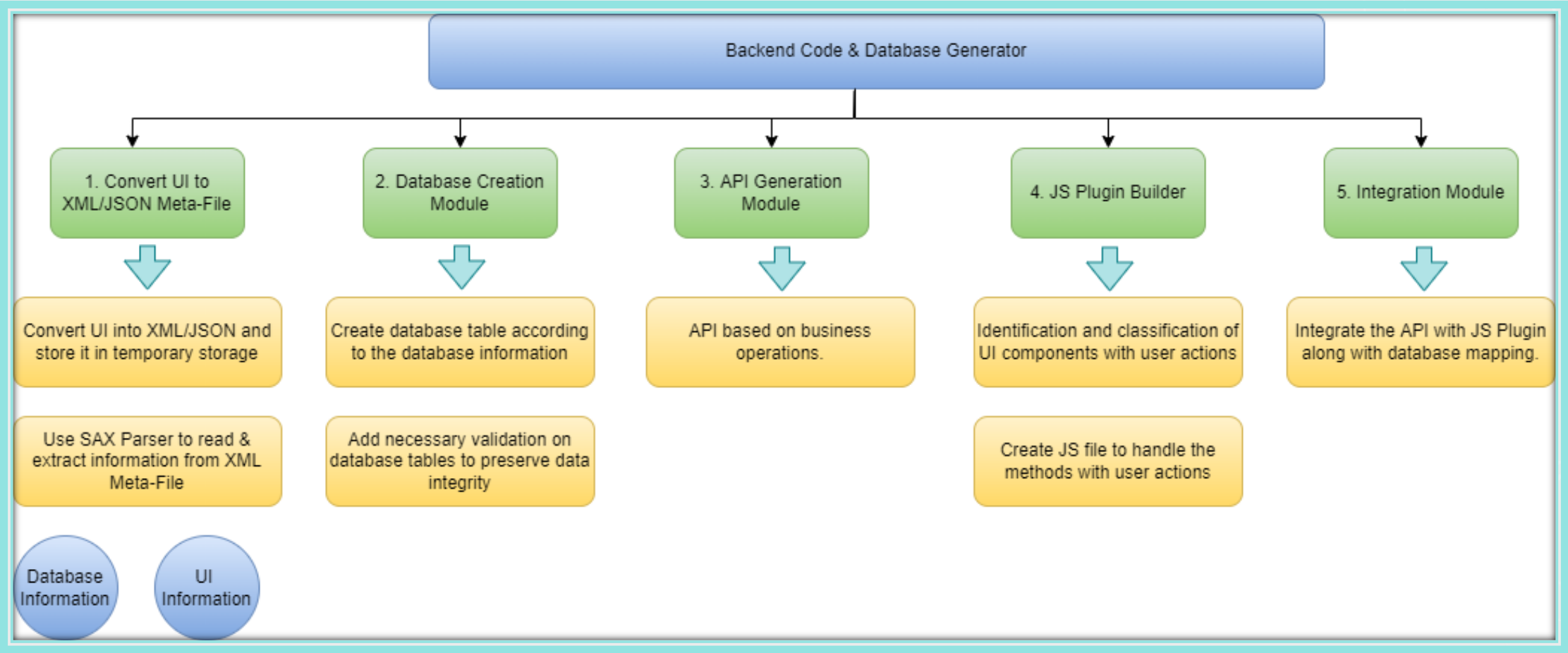
Technologies	Tools
Machine Learning (ML)	Backend server – NodeJS
	Host Backend code – Heroku
	Communication with server – REST API
	Version Management – GitLab
	Project Management – Trello, OneNote



REQUIREMENTS



WORK BREAKDOWN STRUCTURE



GANTT CHART

Task Name	Timeline												
Description	Priority	January	February	March	April	May	June	July	August	September	October	November	December
Project Initiation													
Topic Evaluation	HIGH												
Project Charter	HIGH												
Project Cover sheet	MEDIUM												
Project Proposal (Draft)	MEDIUM												
Conduct a online survey - PHASE 01	HIGH												
Project proposal presentation	HIGH												
Project Proposal - Final report	HIGH												
Organizing and Requirement Anal													
System planning	HIGH												
Collecting required data - PHASE 02	MEDIUM												
Technology & algorithm analysis and selection	HIGH												
Writing Research paper	HIGH												
Implementation													
Development of UI to XML/JSON Converter	HIGH												
Database creation model	HIGH												
API geenration module	HIGH												
JS Plugin Builder	HIGH												
Final Integration Module	HIGH												
Testing													
Developer Testing	HIGH												
User-acceptance testing	MEDIUM												
Final Report	HIGH												
Final Evaluation	HIGH												



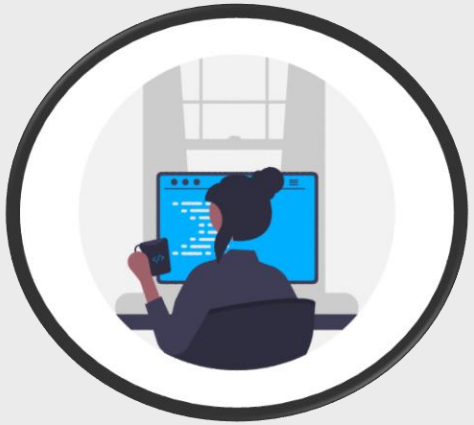
IT19144986|H.M. KASUNI NAVODYA

BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY

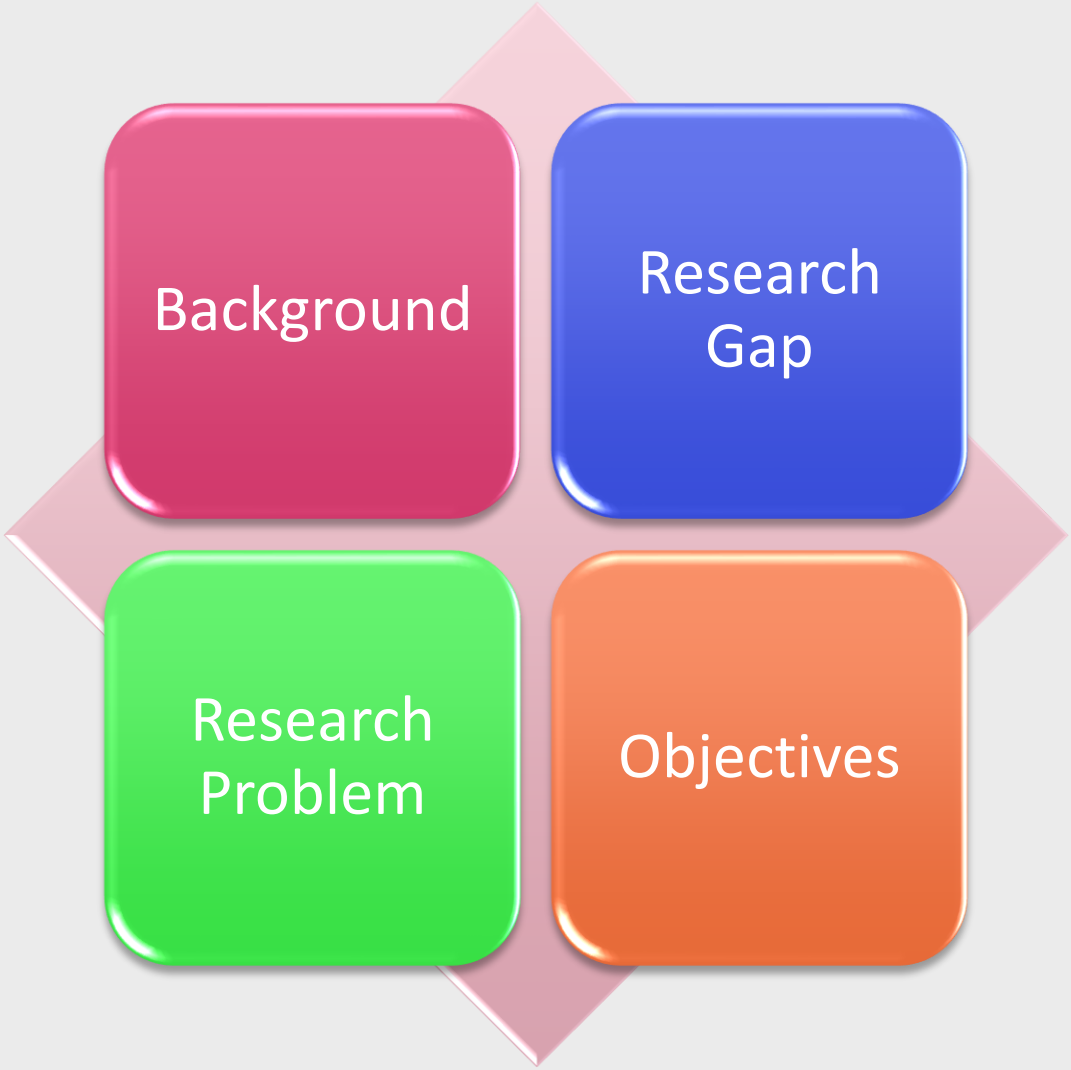
(SPECIALIZATION IN SOFTWARE ENGINEERING)

RESEARCH COMPONENT

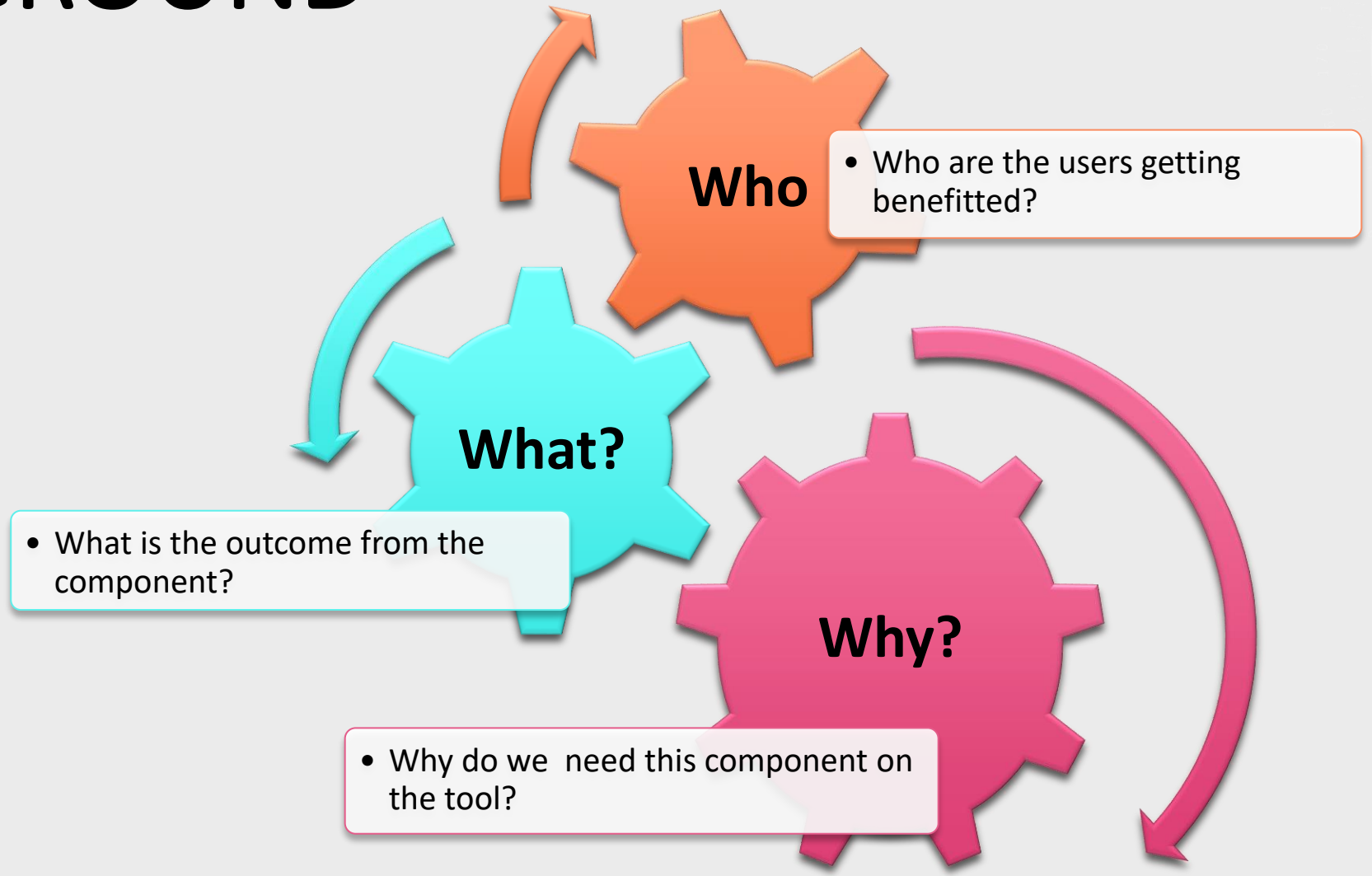
FRONT-END CODE GENERATION FOR WEB APPLICATIONS USING LOGIC NODE CONCEPT



INTRODUCTION



BACKGROUND


Who
• Who are the users getting benefitted?
What?
• What is the outcome from the component?
Why?
• Why do we need this component on the tool?

RESEARCH GAP

Features	Pix2Code [11]	Sketch2Code [12]	UJECTOR [13]	RPB Tool [14]	'UI Builder'
Performance & efficiency of frontend	✗	✗	✗	✗	✓
Accuracy of generated code	✓	✓	✓	✗	✓
Attractive & user-friendly interfaces	✗	✗	✓	✗	✓
Usage of BL concepts	✗	✗	✗	✓	✓
Coding standards & best practices	✓	✗	✓	✗	✓
Support for user preferred technologies	✗	✗	✗	✗	✓
Usability	✗	✗	✗	✗	✓

RESEARCH PROBLEM



How to generate the web frontend code using logic nodes?



How to increase the performance and efficiency of an automated frontend generator?



How to implement frontend code according to user preferred technologies?

OBJECTIVES

Main Objective

❖ Generate an efficient, simple and accurate automated frontend code for web applications.

Sub Objective 1

• Logic Node concepts & BL embedded code generation.

Sub Objective 2

• Develop a tool that follows coding standards and best practices correctly.

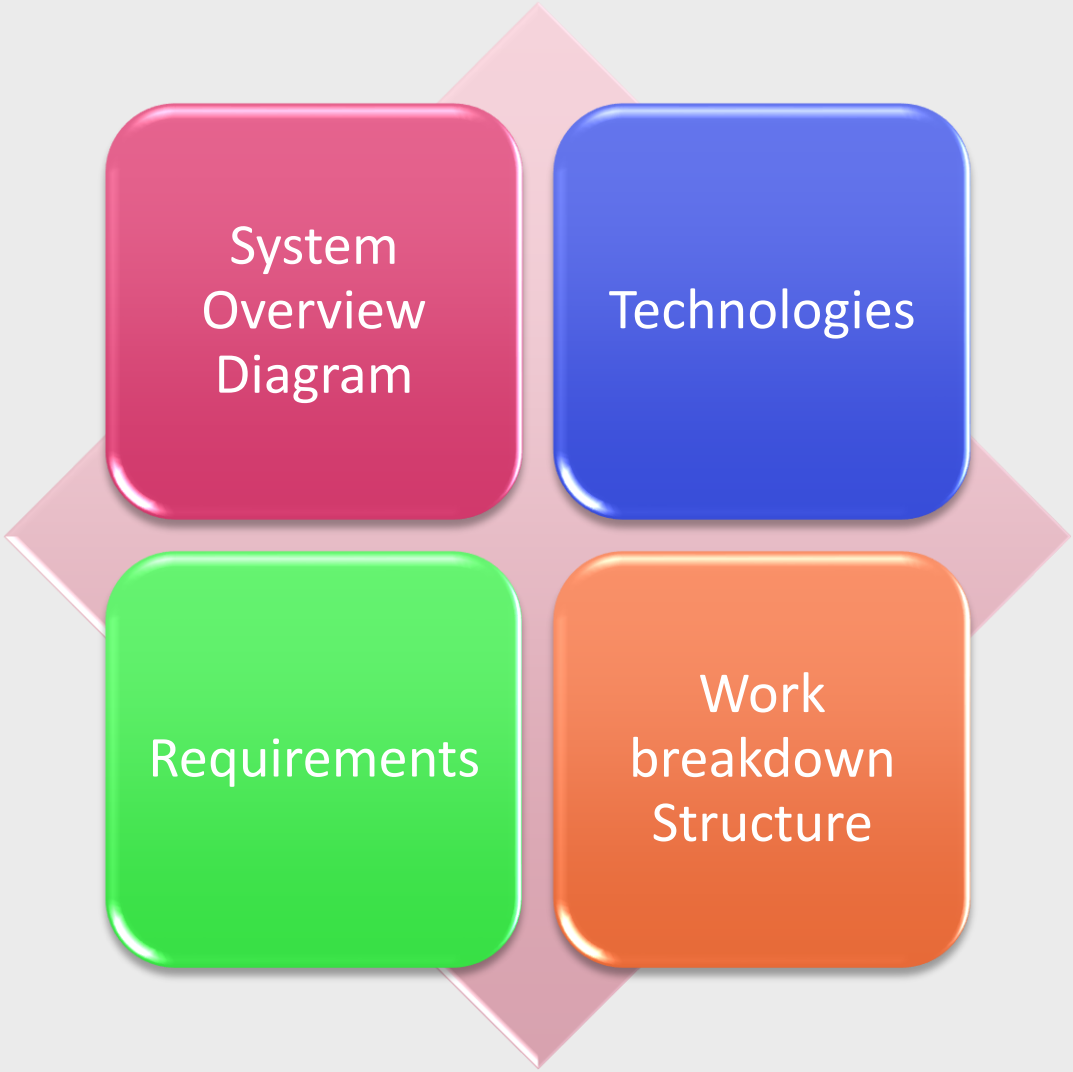
Sub Objective 3

• Provide a user friendly and attractive frontend tool.

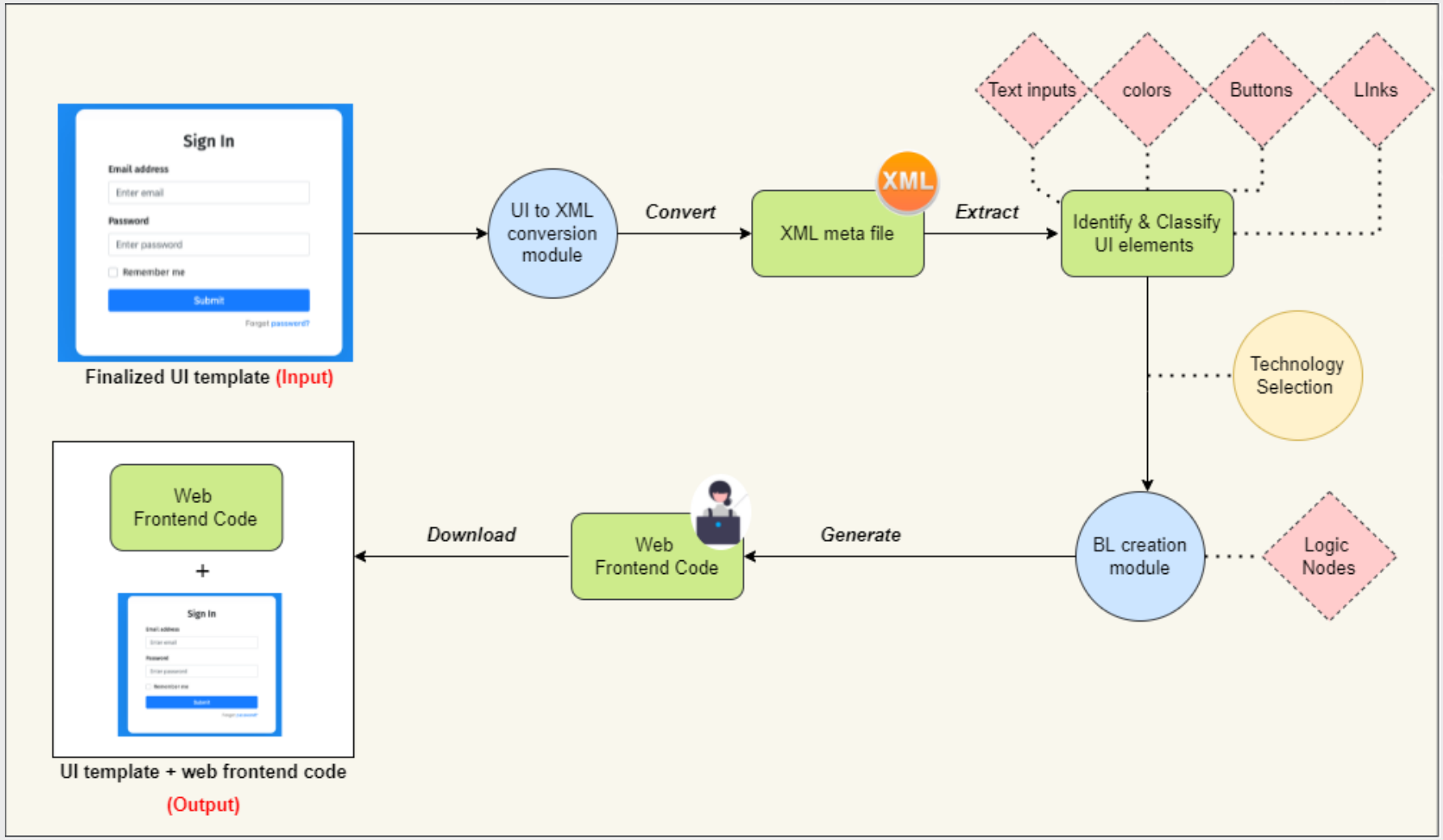
Sub Objective 4

• Increase the productivity in the development process.

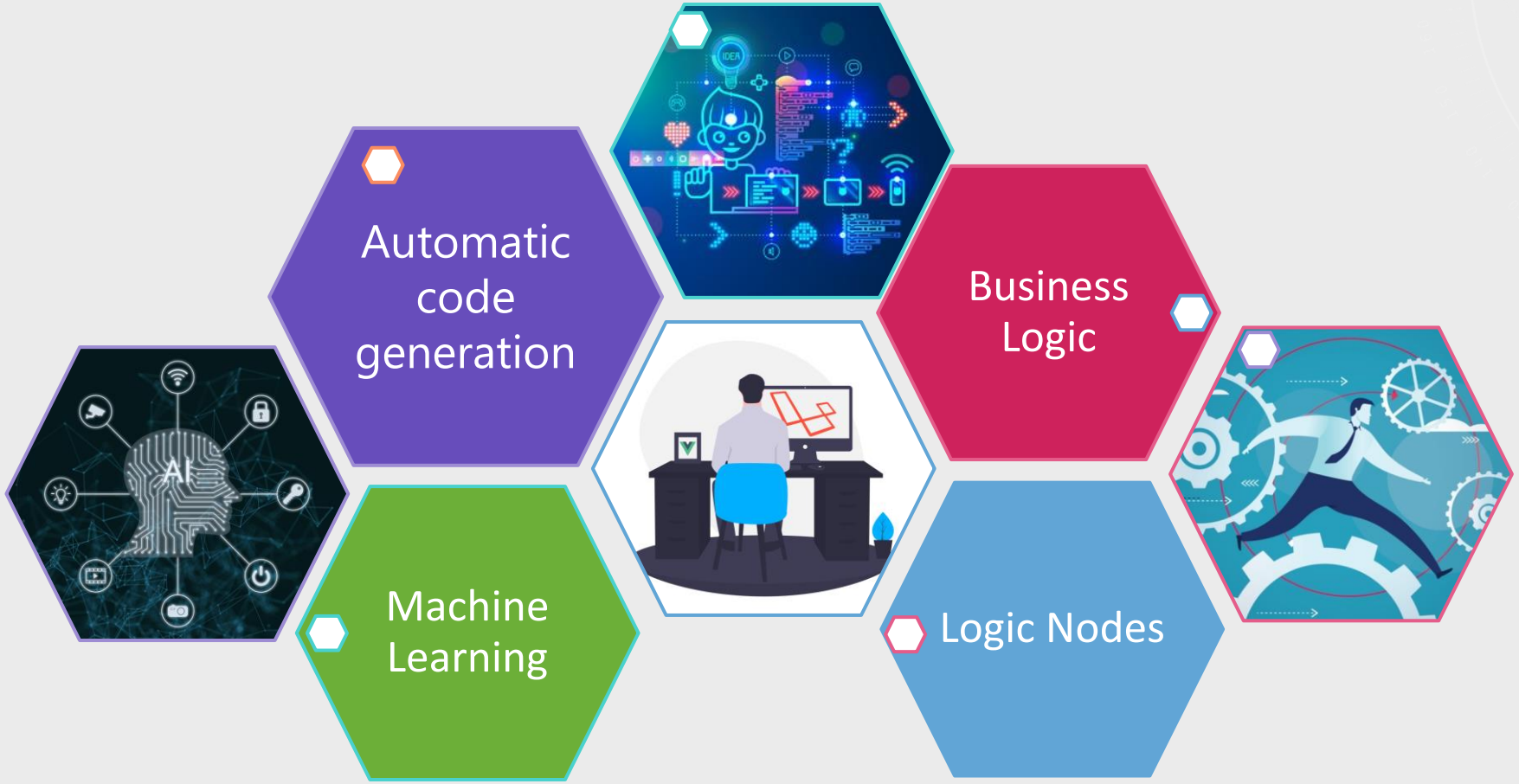
METHODOLOGY



SYSTEM OVERVIEW DIAGRAM



KEY PILLARS


Automatic code generation
Business Logic
Machine Learning
Logic Nodes
AI

TOOLS & TECHNOLOGIES

Technologies

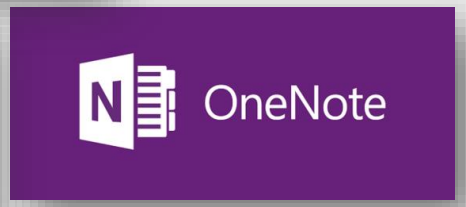
- Machine Learning (ML)







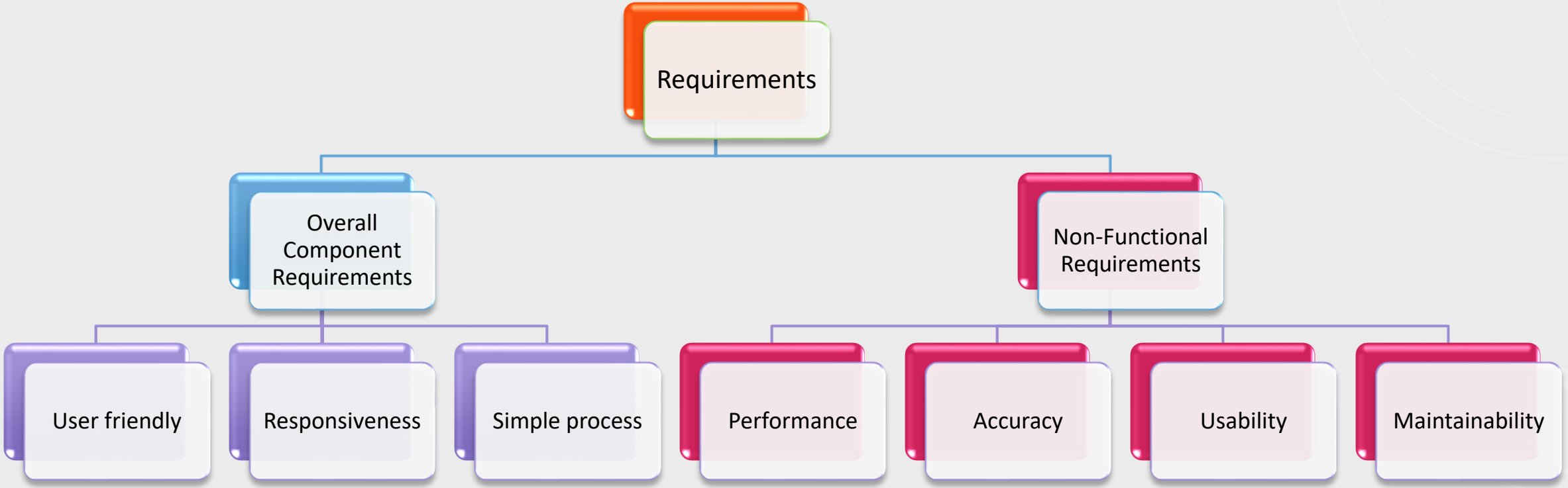




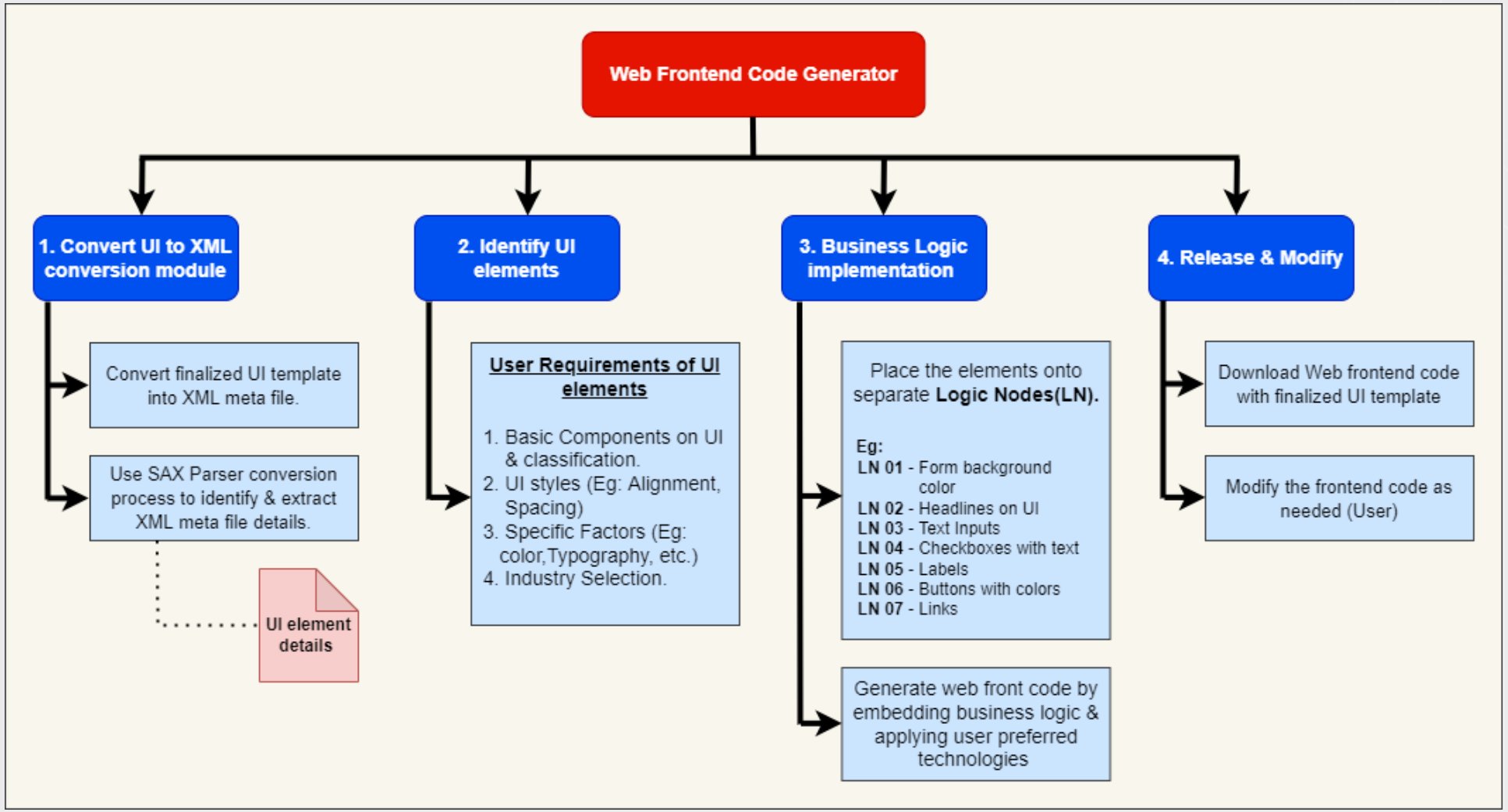
Tools

- Frontend Development – Visual Studio
- Version Management – GitLab
- Project Management – Trello, OneNote

REQUIREMENTS



WORK BREAKDOWN STRUCTURE



GANTT CHART

Task Name	Timeline												
Description	Priority	January	February	March	April	May	June	July	August	September	October	November	December
Project Initiation													
Topic Evaluation	HIGH												
Project Charter	HIGH												
Project Cover sheet	MEDIUM												
Project Proposal (Draft)	MEDIUM												
Conduct a online survey - PHASE 01	HIGH												
Project proposal presentation	HIGH												
Project Proposal - Final report	HIGH												
Organizing and Requirement Anal													
System planning	HIGH												
Collecting required data - PHASE 02	MEDIUM												
Technology & algorithm analysis and selection	HIGH												
Writing Research paper	HIGH												
Implementation													
Template UI to XML Conversion.	HIGH												
Extract XML Meta-file to identify UI elements.	HIGH												
Place the UI elements onto separate logic nodes.	HIGH												
Using Logic Node for BL embedded frontend code generation.	HIGH												
Testing													
Testing	HIGH												
Final Report	HIGH												
Final Evaluation	HIGH												



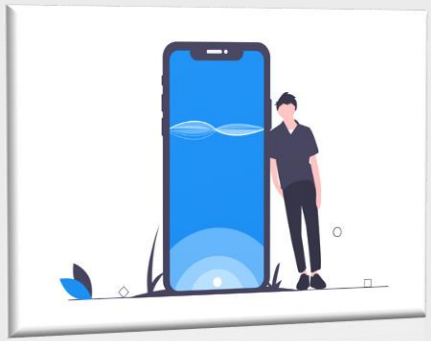
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BSC (HONS) DEGREE IN INFORMATION TECHNOLOGY

(SPECIALIZATION IN SOFTWARE ENGINEERING)

RESEARCH COMPONENT

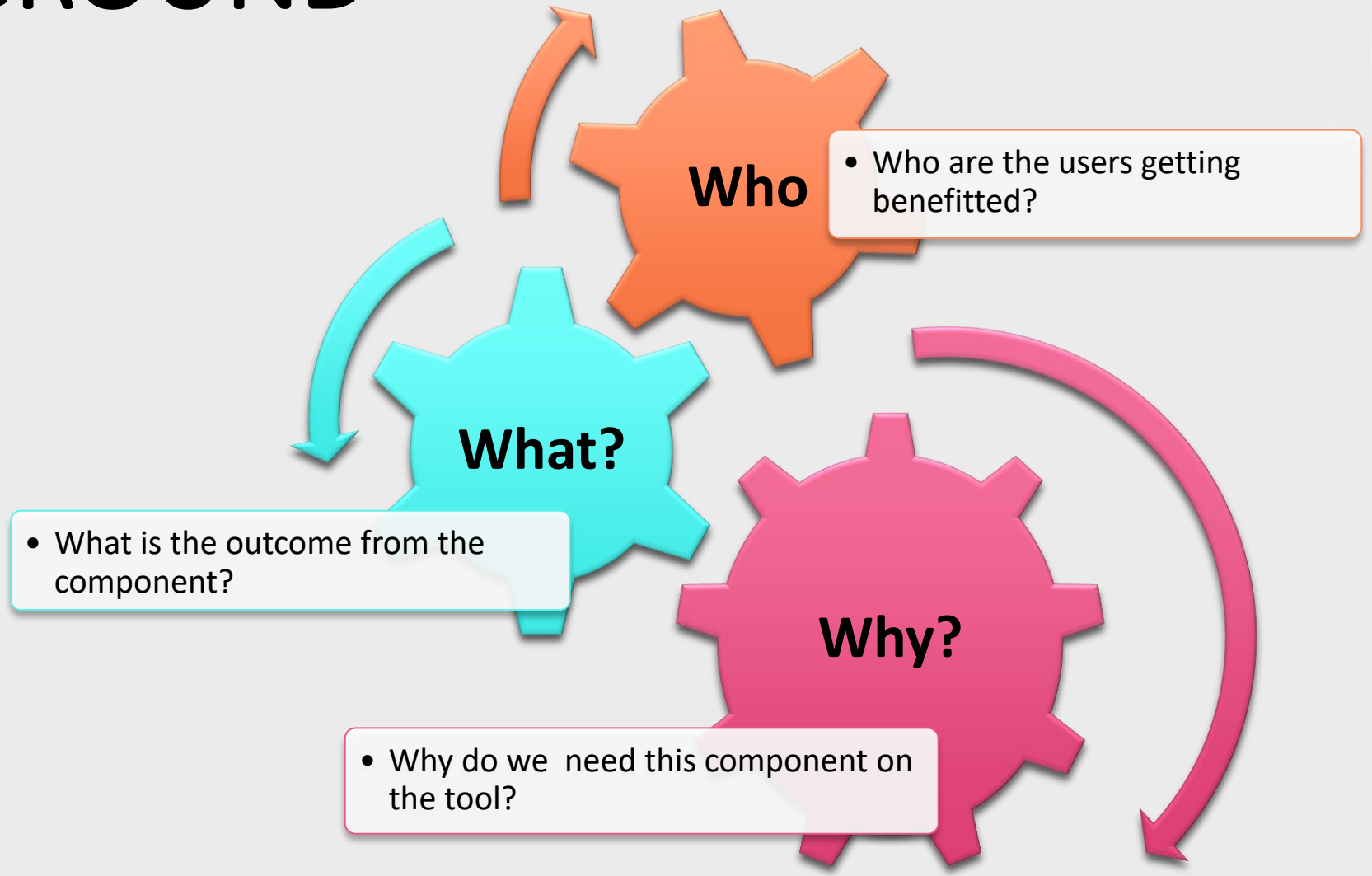
FRONT-END CODE GENERATION FOR MOBILE APPLICATIONS USING MODEL DRIVEN ARCHITECTURE



INTRODUCTION



BACKGROUND


Who
• Who are the users getting benefitted?
What?
• What is the outcome from the component?
Why?
• Why do we need this component on the tool?

RESEARCH GAP

Features	[15]	[16]	[17]	'UI Builder'
Suitable for iOS	✓	✓	✗	✓
Suitable for Android OS	✓	✗	✓	✓
Suitable for Windows OS	✓	✗	✓	✓
Suitable for cross-platform	✓	✗	✗	✓
Responsiveness	✗	✓	✗	✓
Usability	✗	✓	✗	✓

[15] – IFML: The Interaction Flow Modeling Language
[16] – Behrens, Heiko. "MDSD for the iPhone: developing a domain-specific language and IDE tooling to produce real world applications for mobile devices." *Proceedings of the ACM international conference companion on Object oriented programming systems languages and applications companion*. 2010.
[17] – Parada, Abilio, Milena Marques, and Lisane B. de Brisolara. "Automating mobile application development: UML-based code generation for Android and Windows Phone." *Revista de Informática Teórica e Aplicada* 22.2 (2015): 31-50.

RESEARCH PROBLEM



How to generate the mobile frontend code for cross-platform applications?



How to manage the responsiveness of the code for multiple devices?

OBJECTIVES

Main Objective

❖ Generate the mobile frontend code for cross-platform applications

Sub Objective 1

- Responsiveness code for multi devices.

Sub Objective 2

- Provide a better UI front end code generator tool.

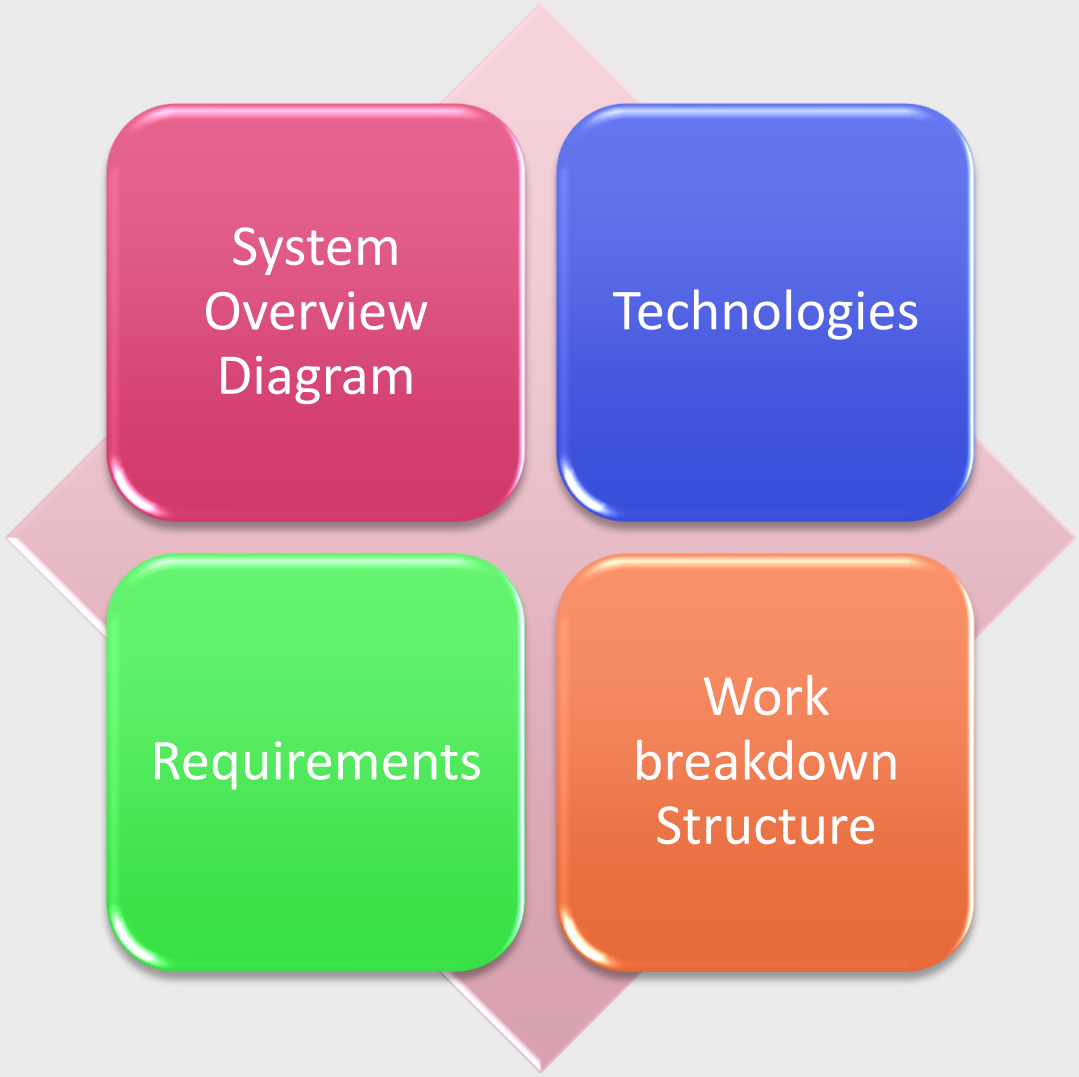
Sub Objective 3

- Make the tool with high performance.

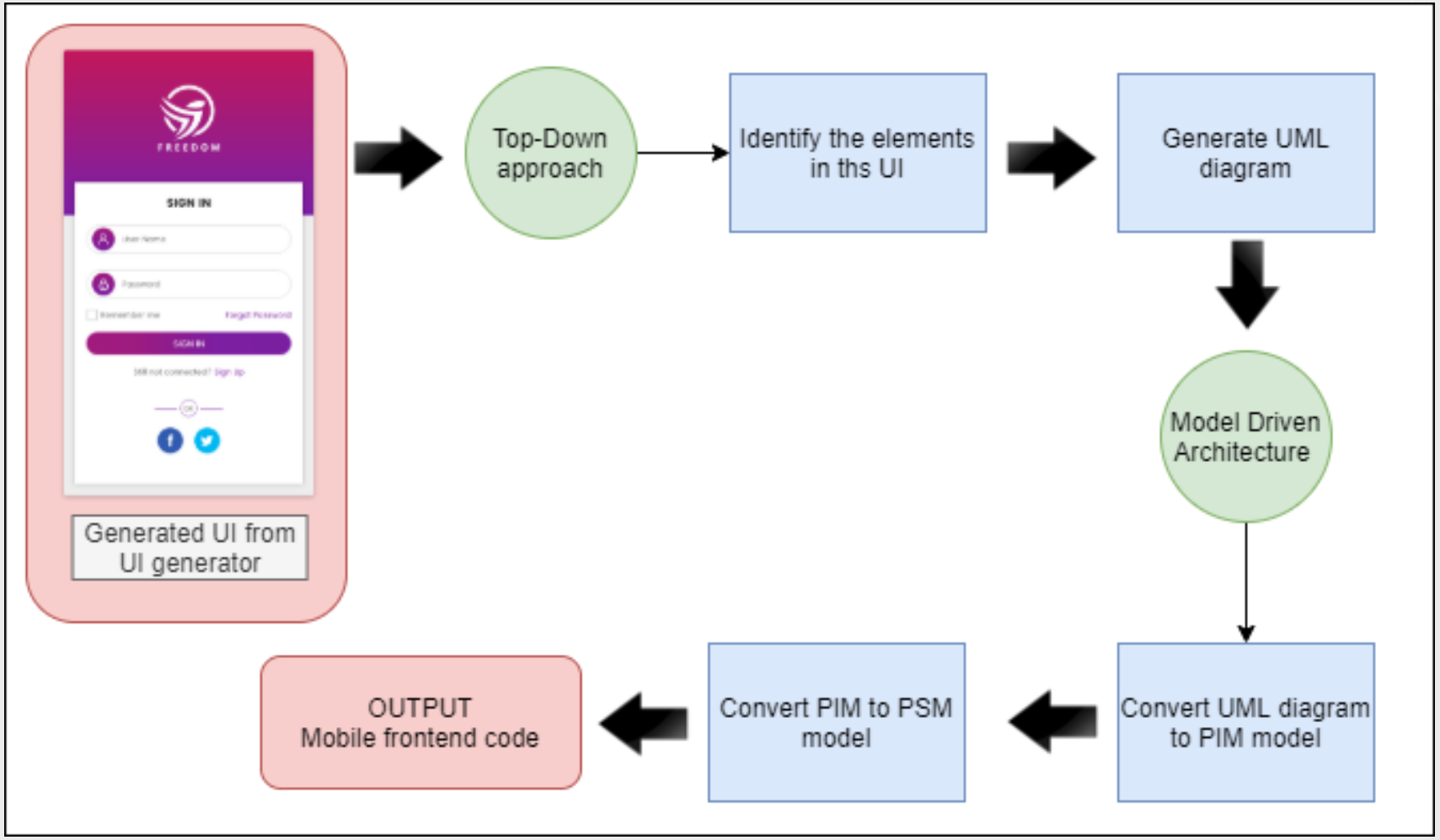
Sub Objective 4

- Implement a user-friendly tool.

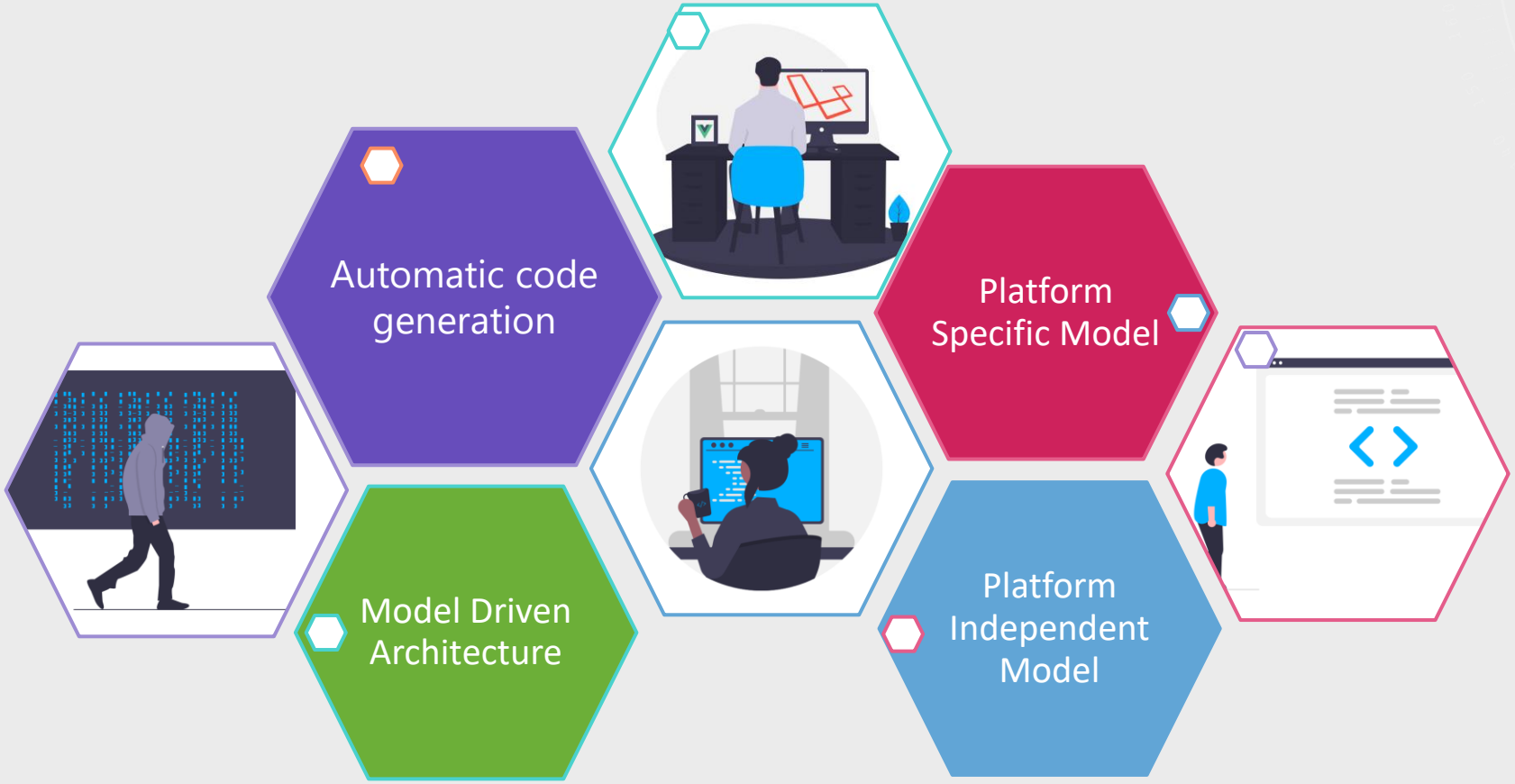
METHODOLOGY



SYSTEM OVERVIEW DIAGRAM

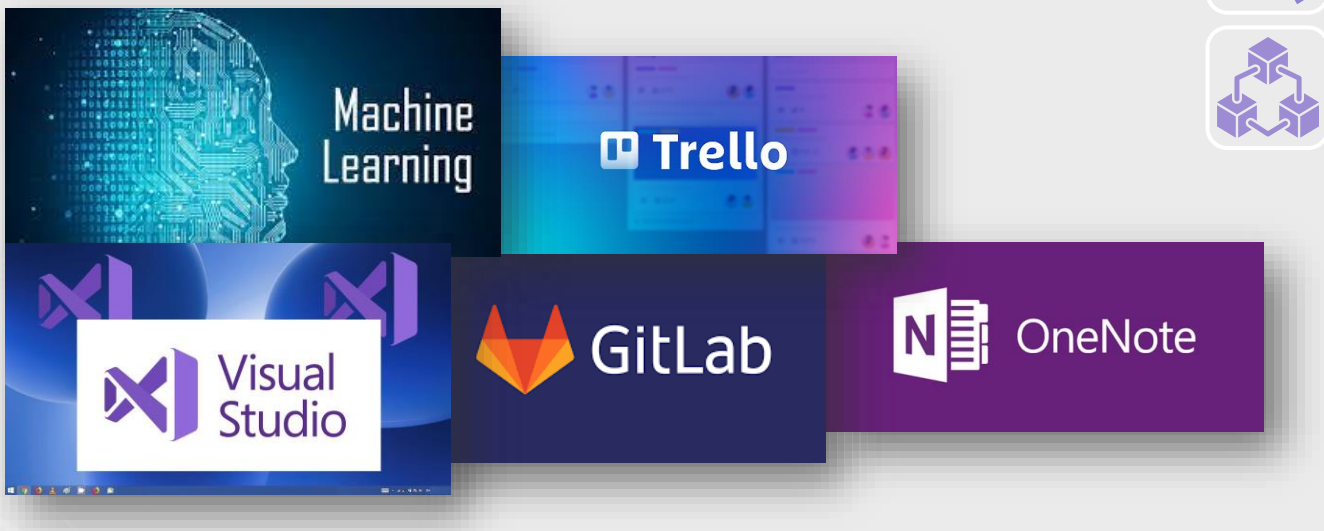


KEY PILLARS

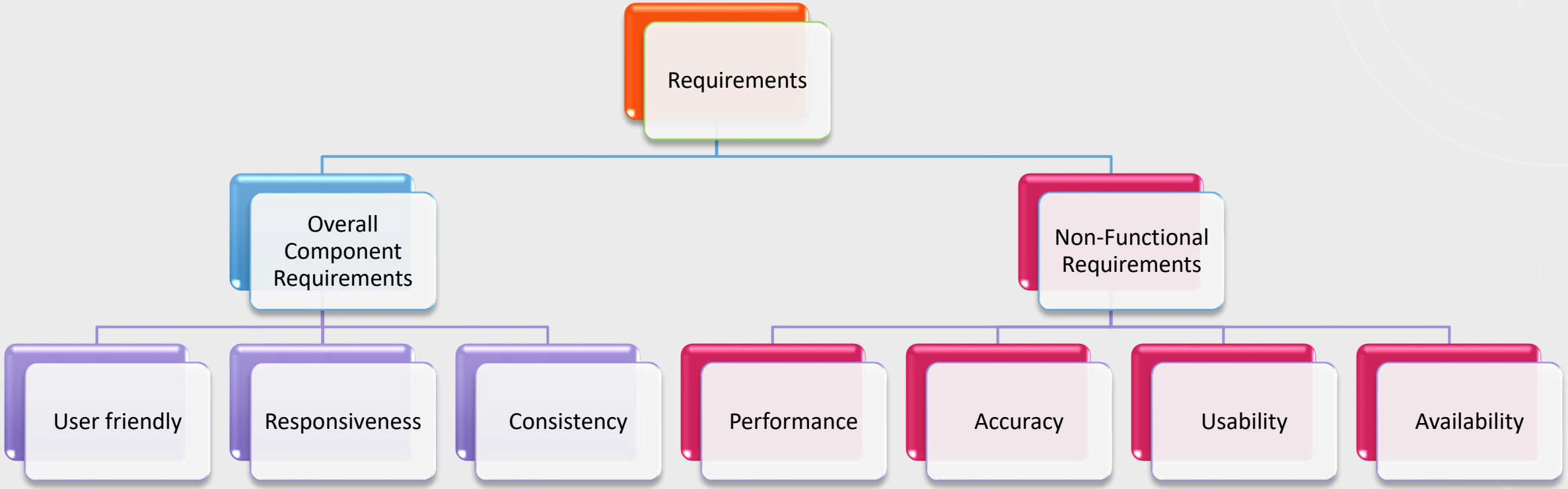

Automatic code generation
Platform Specific Model
Model Driven Architecture
Platform Independent Model

TOOLS & TECHNOLOGIES

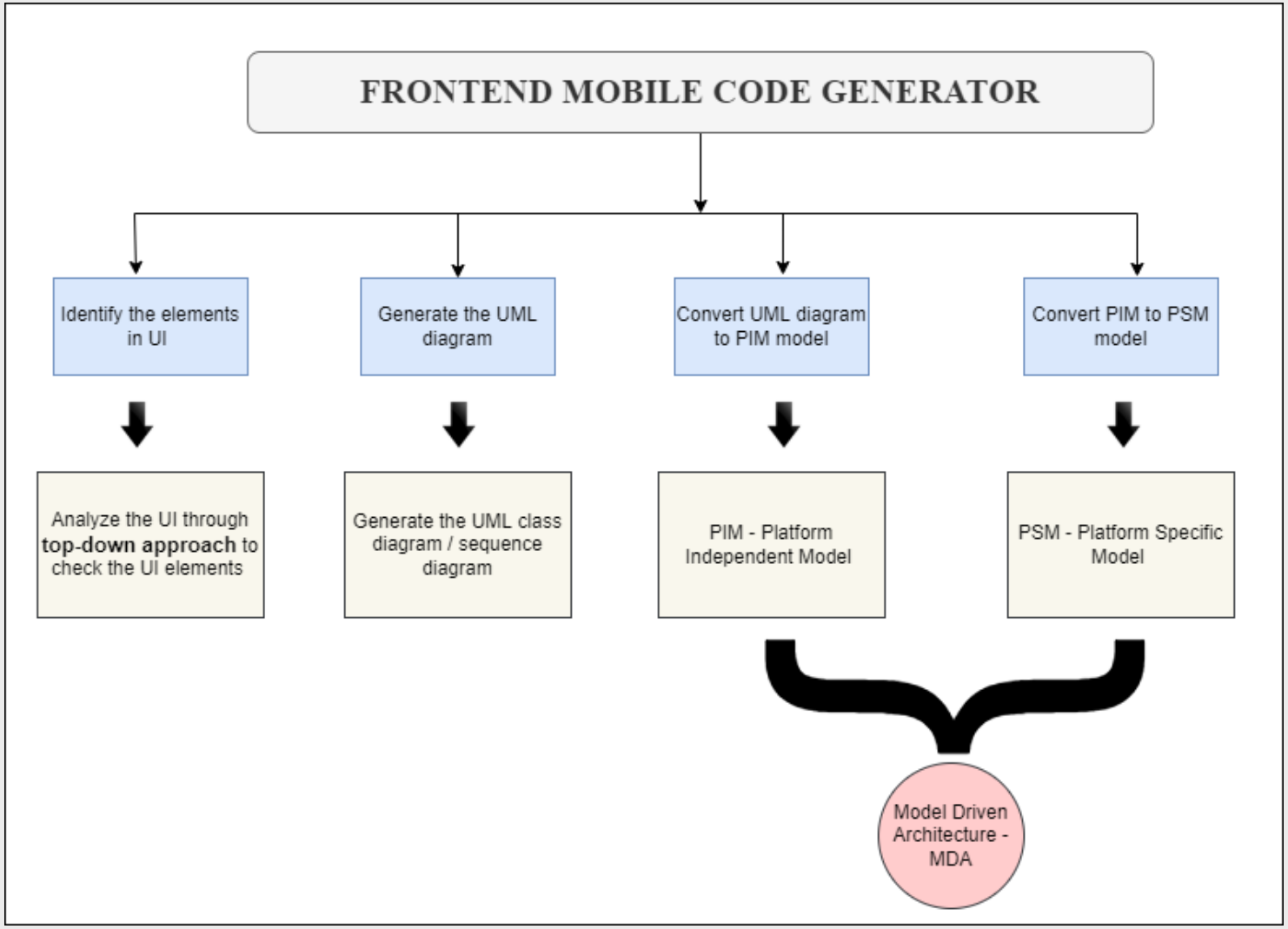
Technologies	Tools
Machine Learning (ML)	Frontend Development – Visual Studio
	Version Management – GitLab
	Project Management – Trello, OneNote



REQUIREMENTS



WORK BREAKDOWN STRUCTURE



GANTT CHART

Task Name	Timeline												
Description	Priority	January	February	March	April	May	June	July	August	September	October	November	December
Project Initiation													
Topic Evaluation	HIGH												
Project Charter	HIGH												
Project Cover sheet	MEDIUM												
Project Proposal (Draft)	MEDIUM												
Conduct a online survey - PHASE 01	HIGH												
Project proposal presentation	HIGH												
Project Proposal - Final report	HIGH												
Organizing and Requirement Anal													
System planning	HIGH												
Collecting required data - PHASE 02	MEDIUM												
Technology & algorithm analysis and selection	HIGH												
Writing Research paper	HIGH												
Implementation													
Identifying UI elements.	HIGH												
Generate UML diagram.	HIGH												
Convert UML to PIM Model.	HIGH												
Convert PIM to PSM Model.	HIGH												
Testing													
Developer Testing	HIGH												
User Acceptance Testing	HIGH												
Final Report	HIGH												
Final Evaluation	HIGH												

COMMERCIALIZATION

- Developers being the targeted users of our solution.
- Our tools helps developers from startups and organizations that involve in developing social software products to reduce their development time and enhance their process.
- The approach used in our tool is a single technology where both web and mobile development teams will be benefitted.
- Licensing schemes targeting individual software engineers, UI designers, frontend and backend developers.
- Separate flexible licensing schemes targeting software companies.
- Standard, professional and enterprise level versions of the solution available.



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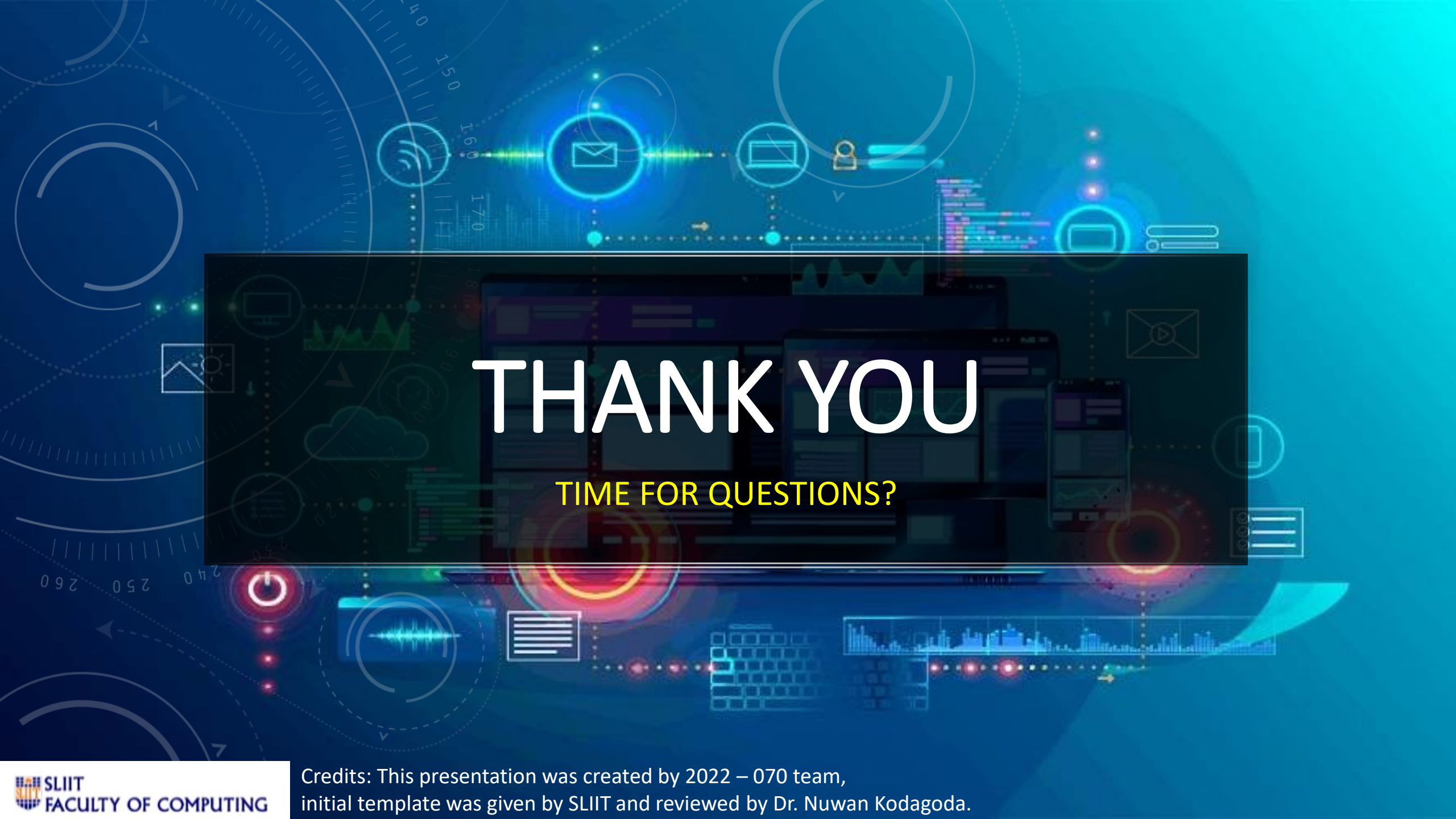
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